

Lorenz Energy Cycle Climatology for the Southwestern Atlantic Cyclones

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Abstract

This study presents a climatological assessment of the Lorenz Energy Cycle (LEC) applied to South Atlantic cyclones, using a Semi-Lagrangian framework. The published analysis included 6,789 cyclones identified from ERA5 reanalysis (1979–2020); the validated corrected rerun currently comprises 3,820 of those systems, with LEC components averaged across four objectively defined life cycle phases: incipient, intensification, mature, and decay. The corrected results show that baroclinic conversion is the dominant conversion pathway, especially during intensification. Barotropic conversion is substantially weaker than reported: it is close to neutral in the pooled median, changes sign across phases and cases, and does not reach magnitudes 2–3 times larger than its baroclinic counterpart. Diabatic generation of eddy available potential energy plays a secondary but relevant role, particularly during intensification. Eddy kinetic energy imports are most significant during the early phases, reinforcing development. Despite substantial variability among systems, an EOF analysis shows a recognizable leading energy structure, whereas higher modes can exchange rank after pattern matching. The leading modes remain linked to differences in cyclone intensity,

genesis region, and seasonality. Among the most intense systems, five corrected clusters retain differing energetic configurations, although their membership and composition change. These findings demonstrate that South Atlantic cyclones encompass a spectrum of dynamical behaviors and can be classified based on their energy cycle characteristics. This study provides a large-sample application of the Semi-Lagrangian LEC to extratropical cyclones in the South Atlantic and highlights the dominant baroclinic pathway together with the case-dependent role of barotropic conversion.

Keywords: Cyclone energetics, Baroclinic conversion, Barotropic conversion, Latent heat release, South Atlantic cyclones

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1 Introduction

Cyclones play a crucial role in the Earth’s climate system by redistributing heat, moisture, and momentum across different regions. These dynamic weather systems, contribute to the global energy balance by transferring energy from the tropics to higher latitudes. Understanding cyclone behavior is essential for improving climate projections and assessing future climate change impacts. In South America, surface cyclones influence precipitation regimes across the continent (Reboita et al., 2010, 2018; de Souza et al., 2022) often contributing to extreme rainfall events (e.g., de Souza and da Silva, 2021; de Souza et al., 2024), and are associated with severe meteorological hazards such as intense winds (Cardoso et al., 2022; de Souza and da Silva, 2021), high sea waves (da Silva et al., 2025; Gramscianinov et al., 2023), and storm surges along coastal areas (Leal et al., 2023; Tecchio et al., 2024). These extreme events can have profound socioeconomic impacts, including damage to coastal infrastructure, disruptions to port operations, and interruptions in oil and gas exploration. This is particularly critical in the southeastern region of South America, where large metropolitan areas, such São Paulo, Rio de Janeiro and Buenos Aires are located, as well as the Port of Santos, the largest in Latin America.

The Lorenz Energy Cycle (LEC) is a valuable framework for understanding the conversion and flow of energy within the atmosphere, particularly in relation to synoptic and large-scale processes (Lorenz, 1967). It divides the atmospheric energy budget into four main reservoirs: kinetic and available potential energy, both for zonal (mean

flow) and eddy components. Lorenz (1955) formulated the concept of available potential energy (APE), building upon Margules (1903) concept of energy based on the hypothetical adiabatic redistribution of atmospheric mass. The LEC serves as a foundational tool for quantifying and understanding atmospheric dynamics, by tracking energy transformations and explaining how these forms of energy are generated by adiabatic processes and dissipated by friction.

Substantial efforts have been made to estimate the LEC components for the global circulation (e.g., Muench, 1965; Wiin-Nielsen, 1968; Hu et al., 2004; Li et al., 2007; Oort, 2018), to assess its projected changes under climate change scenarios (e.g., Hernández-Deckers and von Storch, 2010; Veiga and Ambrizzi, 2013; Pan et al., 2017; Michaelides, 2021), and to explore its relationship with phases of the El Niño–Southern Oscillation (Gutierrez et al., 2009). The LEC framework has also been applied in a range of contexts, from idealized baroclinic wave simulations (Kirshbaum et al., 2018; Rantanen et al., 2019), to rotating annulus experiments (Young, 2014), and even to the atmospheres of other planets (e.g., Read et al., 2020). In contrast, studies examining the LEC in relation to cyclonic systems are more limited and primarily consist of case studies. These have typically focused on tropical (Brennan and Vincent, 1980; Veiga et al., 2008), subtropical (Michaelides, 1987; Dias Pinto et al., 2013; Pezza et al., 2014; Cavicchia et al., 2018), and extratropical cyclones (Michaelides, 1992; Wahab et al., 2002; Bulic, 2006; Pezza et al., 2010; Dias Pinto and Rocha, 2011). Noteworthy contributions include the work of Black and Pezza (2013), which investigated explosive cyclones across all major ocean basins over a 32-year period and identified a universal signature of explosive cyclogenesis, and Okajima et al. (2021), which quantified the contribution of migratory cyclones to global energetics, highlighting their role in accelerating westerly winds worldwide.

Although the original formulation by Lorenz (1967) was intended for global-scale studies, analyzing specific atmospheric regions is valuable for understanding the mechanisms driving the development of individual systems, such as cyclones, and the energy conversions throughout their life cycle. The initial framework for such regional analysis was introduced by Muench (1965), who studied Northern Hemisphere stratospheric circulation during winter. Subsequent formulations by Dutton and Johnson (1967), Vincent and Chang (1973), and Smith (1980) refined the approach, but it was only with Brennan and Vincent (1980) that a generalized set of equations, including both eddy and zonal components of kinetic and APE for limited regions in the troposphere, was established. However, cyclones are mobile systems that often traverse large regions (Hoskins and Hodges, 2002; de Souza et al., 2024, e.g.). Since these studies employ the LEC within an Eulerian framework, the computational domains required to analyze cyclone energetics can become quite large (Black and Pezza, 2013, e.g.), potentially incorporating other atmospheric features (e.g., migratory anticyclones, upper-level troughs, large convective systems), which limits the interpretation of results.

In this context, Michaelides et al. (1999) developed a Semi-Lagrangian framework for analyzing cyclone energetics, where the computational domain is movable. This approach minimizes the impact of neighboring circulations on the system’s energy dynamics by employing a regional computational domain that follows the cyclone

116 track. In the present study, we present a LEC climatology applied to cyclonic sys-
 117 tems using such a framework, as well as the first LEC climatology (considering both
 118 Eulerian and Semi-Lagrangian frameworks) for cyclones in the Southwestern Atlantic.
 119 Over 6,700 cyclones with genesis in the Southwestern Atlantic region are analyzed,
 120 and their energy cycles are divided into distinct life cycle phases: incipient, intensifica-
 121 tion, mature, and decay. This division allows for a more comprehensive understanding
 122 of the energy flows and the dynamical mechanisms acting across each phase of the
 123 cyclone’s life cycle.

124 2 Materials and methods

125 2.1 Data

126 For both the cyclone tracking and energetics computations, we employed the fifth-
 127 generation reanalysis from the European Centre for Medium-Range Weather Forecasts
 128 (ECMWF), known as ERA5 (Hersbach et al., 2020). ERA5 data are produced using
 129 the Integrated Forecasting System (IFS) Cy41r2, which assimilates a wide range of
 130 observations, including satellite and in-situ measurements. It provides global coverage
 131 of numerous atmospheric, ocean-wave, and land-surface parameters at a horizontal
 132 resolution of approximately 31 km (0.25°). Data were requested at all 37 pressure
 133 levels, ranging from 1000 hPa to 1 hPa, with a temporal interval of three hours; the
 134 verified toolkit analysis used the 32 levels from 1000 hPa to 10 hPa.

135 The cyclone tracks used in this study were obtained from the "Atlantic Extratrop-
 136 ical Cyclone Tracks Database" (Gramscianinov et al., 2020). This database tracks the
 137 central relative vorticity of cyclones at the 850 hPa level (ζ_{850}) using the TRACK algo-
 138 rithm (Hodges, 1994, 1995) and the method outlined by Hoskins and Hodges (2002).
 139 This algorithm has been employed in previous studies to assess cyclone climatologies
 140 in the South Atlantic region (Gramscianinov et al., 2019, 2020; de Souza et al., 2024).
 141 The database covers the period from 1979 to 2020 and spans the entire Atlantic Ocean
 142 within the spatial domain of 15°S – 55°S and 75°W – 20°E . The use of ERA5, instead of
 143 other reanalysis datasets, is justified by its higher spatial resolution, which improves
 144 cyclone detection in regions with complex orography and temperature gradients, such
 145 as the SESA region (Gramscianinov et al., 2020). A detailed description of the tracking
 146 methodology can be found in Gramscianinov et al. (2020).

147 The cyclone tracking procedure employed multiple criteria to ensure accurate
 148 cyclone detection. These criteria required cyclones to have a duration of at least
 149 24 hours and a minimum displacement of 1000 km, consistent with previous South
 150 Atlantic cyclone climatologies (Sinclair, 1995; Gramscianinov et al., 2019). Systems
 151 that spent over 80% of their life cycle over continental regions were excluded from
 152 the analysis to avoid counting thermal lows and lee troughs (e.g. Crespo et al., 2021).
 153 Notably, although the TRACK algorithm’s calibration and sensitivity are focused on
 154 extratropical cyclones — the majority of systems in this region (Marrafon et al., 2022)
 155 — the methodology does not explicitly exclude subtropical or tropical cyclones.

156 To focus on the Southwestern Atlantic Ocean, only cyclones with genesis near
 157 the South American coast were considered, specifically those originating within the
 158 eastern Argentina (ARG), La Plata River basin (LA-PLATA), and southeast Brazil

(SE-BR) genesis regions (Gramscianinov et al., 2019; de Souza et al., 2024) (Figure 1). After applying these selection criteria, the database initially comprised a total of 7931 cyclones. The spatial distribution of these genesis regions and the cyclone track density for all selected systems are presented in Figure 1. Noteworthy, in traditional cyclone studies, cyclone density is often presented for a single time step or normalized by individual tracks (e.g., Hoskins and Hodges, 2005; Gramscianinov et al., 2019). In contrast, Figure 1 displays cyclone density computed across all time steps within a 5° spherical cap. Cyclones may form within the cap, move into it from adjacent regions, remain for several time steps, or pass through it entirely. Each of these time-step occurrences is counted, meaning that the density values in the figure reflect multiple temporal passages rather than unique storm events.

Among the original 7,931 cyclones, 4,445 originated in ARG (56.0%), 1,870 in LA-PLATA (23.6%), and 1,616 in SE-BR (20.4%). However, due to data availability issues during the download process from the ERA5 database, not all cyclones could be considered in this study, resulting in a reduced total of 6,789 cyclones. These issues were related to the ERA5 API instability during the second half of 2023 and the first half of 2024, when ECMWF was migrating its data infrastructure and updating its API services. As a result, slow transfers and repeated interruptions caused incomplete or failed downloads, despite multiple attempts. This led to some cyclones being excluded from the analysis. The remaining dataset still includes a sufficiently large number of systems for robust analysis and generalization of the results. Of the 6,789 cyclones, 4,015 originated in ARG (59.1%), 1,288 in LA-PLATA (18.9%), and 1,486 in SE-BR (21.8%). The reduction corresponds to a 14.4% decrease in the number of cyclones, with region-specific reductions of 9.7% for ARG, 31.1% for LA-PLATA, and 8.0% for SE-BR.

The corrected production rerun validated 3,820 of the 6,789 archived cyclones and 15,829 lifecycle rows. Corrected results in this manuscript therefore use the validated 3,820-cyclone population. Comparisons with the full published population combine the toolkit correction with this population change; a correction-only paired control uses the same 3,820 cyclones on both sides.

2.2 Lorenz Energy Cycle Computation

For computing the LEC, we used the the open-source Python application LorenzCycleToolKit (de Souza et al., 2024). The energy budget equations for the zonal and eddy forms of APE and kinetic energy are then expressed as:

$$\frac{\partial A_Z}{\partial t} = BA_Z - C_Z - C_A + G_Z \quad (1)$$

$$\frac{\partial A_E}{\partial t} = BA_E - C_E + C_A + G_E \quad (2)$$

$$\frac{\partial K_Z}{\partial t} = BK_Z + C_Z + C_K + B\Phi_Z + D_Z \quad (3)$$

$$\frac{\partial K_E}{\partial t} = BK_E + C_E - C_K + B\Phi_E + D_E \quad (4)$$

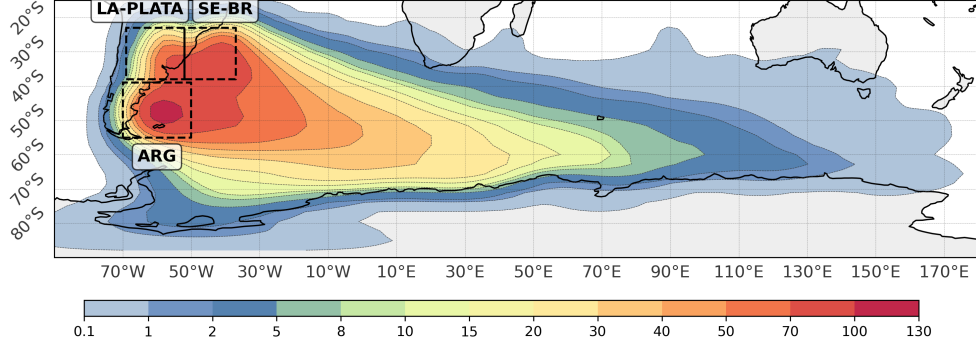


Fig. 1 Cyclone track density for all systems analyzed in this study, highlighting the cyclogenesis regions near the South American coast (ARG, LA-PLATA, and SE-BR). The track density unit is cyclonic centers per 10^6 km^2 per month.

In these equations, the atmospheric energy reservoirs are separated into zonal and eddy components of available potential energy (A_Z , A_E) and kinetic energy (K_Z , K_E). Energy conversions between reservoirs are represented by the following terms: C_A , the conversion between zonal and eddy available potential energy ($A_Z \leftrightarrow A_E$); C_E , the conversion from eddy available potential energy to eddy kinetic energy ($A_E \leftrightarrow K_E$); C_K , the barotropic conversion from eddy kinetic energy to zonal kinetic energy ($K_E \leftrightarrow K_Z$); and C_Z , the conversion from zonal available potential energy to zonal kinetic energy ($A_Z \leftrightarrow K_Z$). Positive C_K transfers energy from K_E to K_Z , whereas negative C_K feeds K_E . The generation of available potential energy (G) and signed frictional contributions to kinetic energy (D) are indicated by subscripts Z and E ; negative D denotes kinetic-energy loss. The boundary flux terms, BA_Z , BA_E , BK_Z , and BK_E , represent the import/export of available potential energy and kinetic energy across the computational domain boundaries. The boundary pressure-work terms $B\Phi_Z$ and $B\Phi_E$ can either supply or remove kinetic energy. The complete mathematical formulation of these terms can be found in the Appendix.

In these formulations, a global reference state is used to define the APE, meaning that the computed APE represents the contribution of local APE to global energetics (Smith, 1980). As a result, it does not account for regional variability, which can introduce bias in the APE computation (see Novak and Tailleux (2018) for a thorough discussion). Despite the limitations of this methodology, the formulations by Muench (1965) and Brennan and Vincent (1980) still provide a valuable framework due to their simplicity, robustness, and ability to yield consistent, interpretable results, enabling direct analysis of the dynamic processes driving cyclones development (Dias Pinto and Rocha, 2011; de Souza et al., 2025, e.g.). Moreover, while using a local APE definition (Novak and Tailleux, 2018, e.g.) offers more precise results, it often comes with higher computational costs and does not provide the same level of direct, practical interpretability.

220 Following the methodologies of Brennan and Vincent (1980); Michaelides (1987);
 221 Veiga et al. (2008); Pezza et al. (2010); Dias Pinto and Rocha (2011), the program
 222 was set to compute the dissipation and generation terms as residuals, defined as:

$$RK_Z = B\Phi_Z + D_Z + \epsilon_{KZ} \quad (5)$$

$$RK_E = B\Phi_E + D_E + \epsilon_{KE} \quad (6)$$

$$RG_Z = G_Z + \epsilon_{GZ} \quad (7)$$

$$RG_E = G_E + \epsilon_{GE} \quad (8)$$

223 Here, RK_Z and RK_E are composite kinetic-energy residuals: they combine bound-
 224 ary pressure work, signed frictional dissipation, unresolved-scale transfers, and accu-
 225 mulated numerical error. Likewise, RG_Z and RG_E combine generation with numerical
 226 error. These residuals are budget-closure terms and must not be interpreted as direct
 227 measurements of a single physical generation or dissipation process. The set of budget
 228 equations becomes:

$$\frac{\partial A_Z}{\partial t} = -C_A - C_Z + RG_Z + BA_Z \quad (9)$$

$$\frac{\partial A_E}{\partial t} = C_A - C_E + RG_E + BA_E \quad (10)$$

$$\frac{\partial K_Z}{\partial t} = C_K + C_Z + BK_Z + RK_Z \quad (11)$$

$$\frac{\partial K_E}{\partial t} = -C_K + C_E + BK_E + RK_E \quad (12)$$

229 A complete depiction of the LEC, with arrows indicating positive energy fluxes,
 230 is presented in Figure 2. This figure also highlights three primary energetic pathways
 231 (baroclinic instability, barotropic conversion, and latent heat release) which will be
 232 discussed throughout the text.

233 Here, we used a Semi-Lagrangian framework (Michaelides et al., 1999), aiming to
 234 minimize interactions with other non-related circulations while capturing the main
 235 structure of the cyclone. A $15^\circ \times 15^\circ$ computational domain centered at the cyclone's
 236 central position, obtained from the TRACK database, was created for each time step.
 237 Given the impracticality of manually selecting an appropriate domain size for each
 238 cyclone in the dataset, a fixed size was employed. The choice of a $15^\circ \times 15^\circ$ computa-
 239 tional domain is justified as it is large enough to capture the effective radius of most
 240 cyclonic systems (Rudeva and Gulev, 2007). The effective cyclone radius is defined as
 241 a measure of cyclone size, determined by establishing a coordinate system centered
 242 on the cyclone and measuring the distance at which the radial pressure gradient first
 243 falls to zero. This domain size also is large enough for accommodating the cyclone's
 244 synoptic structure (e.g. Gramscianinov et al., 2019).

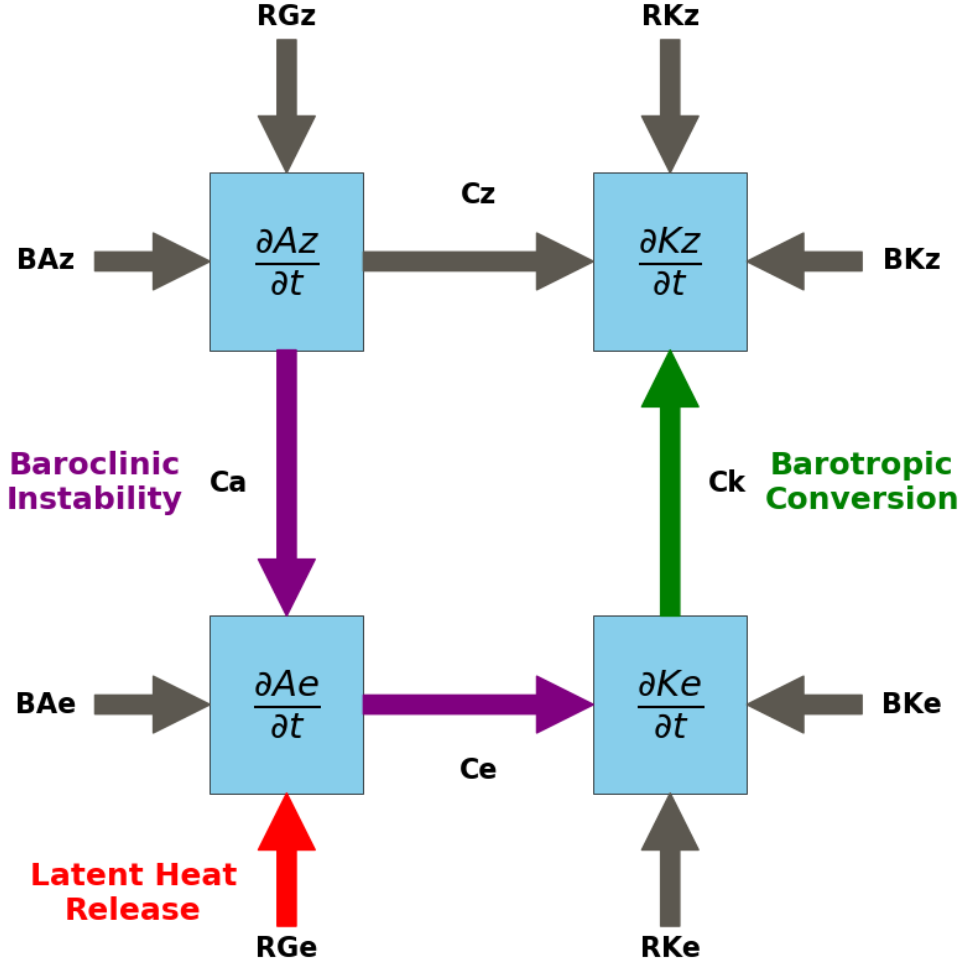


Fig. 2 Representation of the Lorenz Energy Cycle (LEC), with arrows denoting positive energy fluxes. The figure also indicates three main energetic pathways: baroclinic instability (purple), barotropic conversion (green), and latent heat release (red), which will be addressed in detail in the following sections.

2.3 Analysis Methods

A key challenge in computing the LEC for large datasets lies in the variable life durations of cyclones. Averaging energy values across an entire life cycle can obscure distinct dynamical processes tied to different stages of cyclone development, while technical limitations exist in dissecting the life cycle into distinct periods. For example, Black and Pezza (2013) averaged cyclone energetics over periods of 48 hours before explosive cyclogenesis, during explosive deepening, and for 24 and 72 hours after it. However, cyclone life cycles can range from less than 24 hours to over 10 days (Trigo,

2006; Reboita et al., 2010; Gramscianinov et al., 2019), presenting significant physical limitations to such approaches.

To address these issues, we employed Cyclophaser, an open-source Python package designed to detect cyclone life cycle phases (de Souza et al., 2025). This program uses time series of relative vorticity at the system’s central position and its first derivative to identify intensification, mature, and decay phases based on peaks and valleys in the vorticity and its derivative. In this study, we adopted the same life cycle detection approach as de Souza et al. (2024). After computing the LEC for each system, Cyclophaser was used to identify the life cycle phases for all analyzed cyclones, followed by the computation of mean LEC values for each phase. This approach facilitates the investigation of dynamical mechanisms specific to each development phase. For example, distinct energy fluxes are expected to dominate during the intensification and decay phases, particularly near the cyclone center. As shown by de Souza et al. (2024), approximately 60% of the analyzed cyclones follow a classical life cycle, consisting of incipient, intensification, mature, and decay phases, while over 95% display at least one intensification phase. A comprehensive description of the detection methodology, as well as spatial distributions and phase-specific statistics such as duration, displacement, speed, and intensity, can be found in de Souza et al. (2024).

To understand the dominant LEC variability patterns within the TRACK dataset, we performed an Empirical Orthogonal Function (EOF) analysis (Fukuoka, 1951; Lorenz, 1956) using the Python open-source library pyEOF (Zheng, 2021). This analysis reduces the dimensionality of the dataset, retaining the most significant variability, and is particularly useful for uncovering underlying structures and patterns in complex environmental data. Typically, each EOF represents a spatial pattern, while the associated time series, or principal component (PC), describes the temporal evolution of that pattern. However, in this study, each EOF reflects a mode of variability derived from all LEC terms, using their average values computed separately for each cyclone life cycle phase (incipient, intensification, mature, and decay). Consequently, the PCs represent the variation across individual cyclones rather than temporal evolution, and therefore are not displayed individually, as they do not depict any meaningful temporal or spatial feature. Performing EOF analyses separately for each life cycle phase enabled a detailed investigation of the variability in cyclone energetics across distinct stages of cyclone development, as well as throughout the entire energy cycle.

To associate cyclones with predominant EOFs, we first project each cyclone’s energetics onto the PCs obtained from an EOF analysis. This allows us to represent each system in terms of its contribution to the dominant modes of variability. Next, we classify cyclones into EOF(+) and EOF(-) groups, identifying systems where at least one PC reaches extreme values. Specifically, EOF(+) corresponds to cyclones where at least one PC exceeds the 90th percentile (q90), indicating a strong positive projection onto a specific mode. Meanwhile, EOF(-) are cyclones for which at least one PC is below the 10th percentile (q10), indicating strong negative projection onto a specific mode. For each cyclone meeting these criteria, we determine its predominant EOF as the one corresponding to the PC with the highest absolute value. This ensures that each system is classified based on the mode that most strongly characterizes its structure and dynamics.

To determine the LEC patterns of the most intense systems, we first selected only the cyclones in which the maximum central vorticity at 850 hPa exceeded the 90th percentile of the dataset. Subsequently, we applied the K-Means algorithm to the PCs of these intense cyclones to investigate whether distinct energetic characteristics could be identified among them. This clustering analysis was not performed on all cyclones, as sensitivity tests indicated that applying it to the full dataset resulted in groups that differed mainly in intensity rather than exhibiting distinct energetic behaviors (not shown); thus, intensity-based selection was necessary.

K-Means is an unsupervised learning algorithm that partitions data into K clusters by minimizing intra-cluster variance while maximizing separation between groups based on sample similarity (MacQueen et al., 1967; Hartigan and Wong, 1979). The optimal number of clusters was determined using the Elbow Method, which identifies the most suitable number of clusters by plotting the within-cluster sum of squares for different values of K and locating the "elbow point," where the rate of decrease slows, indicating a balance between model complexity and performance. For this analysis, we employed the K-Means implementation from the `scikit-learn` Python package (Hao and Ho, 2019) and used the `yellowbrick` package for the Elbow Method (Bengfort and Bilbro, 2019).

3 Results

3.1 Climatological Features

The exploratory statistical metrics for all LEC terms, presented as averages across the entire cyclone lifecycle, are shown in Table 1, while their probability density functions (PDFs) are provided in Figure 3. Notably, all terms related to the zonal jets (K_Z , BK_Z , and $\frac{\partial K_Z}{\partial t}$) frequently exhibit statistical metrics one order of magnitude higher than their eddy counterparts within the same group (K_E , BK_E , and $\frac{\partial K_E}{\partial t}$), indicating significantly greater energy amount concentrated on the jet streams. Although the energy budgets were computed with the generation and dissipation terms calculated as residuals, the generation terms can be directly computed using ERA5 data. Therefore, their probability density functions (PDFs) are also shown for a more physically based analysis (Figure 3e). Furthermore, the comparison between the directly computed generation values and those estimated by residual computation provides a useful metric for evaluating the accumulation of errors in LEC computation procedure. The dissipation terms were not directly computed in the present study.

All the energy terms (K_z , A_z , A_e , and K_e , Figure 3a) exhibit right-skewed distributions. Among these, A_z shows the largest variability, as reflected in its long-tail distribution, standard deviation (std), interquartile range (IQR), and range (defined as the difference between maximum and minimum values), compared to A_e and K_e , peaking at approximately $4 \times 10^5 \text{ J m}^{-2}$. Conversely, A_e presents the lowest variability, peaking near $1 \times 10^5 \text{ J m}^{-2}$, closer to its mean value. The kinetic energy terms, K_Z and K_E , display similar distributions, with K_Z peaking near $20 \times 10^5 \text{ J m}^{-2}$ and K_E peaking near $2 \times 10^5 \text{ J m}^{-2}$.

The corrected conversion statistics retain the broad variability of C_Z and C_E , which are numerically unchanged by the toolkit correction. The affected terms change

Table 1 Summary Statistics of Lorenz Energetics Components, computed as averages across the entire cyclone lifecycle: mean, median, standard deviation (std), 25th percentile (Q25), 75th percentile (Q75), interquantile range (IQR), and range. The IQR measures the range within which the central 50% of the values fall, while the range is computed as the difference between the maximum and minimum values. The units for the energy terms (A_Z , A_E , K_Z , and K_E) are $10^5 J m^{-2}$ and $W m^{-2}$ for the remaining terms.

Term	Mean	Median	Std Dev	Q25	Q75	IQR	Range
Az	5.55	4.68	3.71	2.77	7.49	4.72	36.46
Ae	1.62	1.24	1.20	0.77	2.09	1.32	11.07
Kz	28.55	26.19	15.10	17.34	37.12	19.78	114.55
Ke	3.70	2.95	2.62	1.92	4.63	2.71	24.81
Cz	0.52	0.38	4.85	-1.96	2.92	4.89	86.70
Ca	0.93	0.46	1.57	0.02	1.44	1.41	19.78
Ck	-3.56	-1.63	9.51	-5.77	0.47	6.24	206.43
Ce	3.84	2.47	4.92	0.63	5.84	5.21	54.74
BAz	3.54	2.03	7.67	-0.67	6.67	7.34	175.05
BAe	0.58	0.16	4.40	-1.13	2.02	3.15	75.72
BKz	-7.93	-5.53	31.75	-23.75	7.83	31.58	445.63
BKe	0.47	0.08	8.59	-2.73	3.22	5.95	146.87
$B\Phi_Z$	63.87	49.25	128.05	-9.12	132.35	141.47	1419.08
$B\Phi_E$	39.66	31.19	126.50	-28.84	109.47	138.32	1353.99
Gz	-0.31	-0.17	2.66	-1.39	0.81	2.21	68.74
Ge	1.17	0.56	3.14	-0.26	2.21	2.46	58.39
$\frac{\partial A_z}{\partial t}$	-0.07	-0.14	4.46	-2.04	1.68	3.71	106.39
$\frac{\partial A_e}{\partial t}$	-0.23	-0.10	1.92	-0.85	0.51	1.35	48.19
$\frac{\partial K_z}{\partial t}$	1.59	0.75	13.38	-5.33	8.15	13.48	251.24
$\frac{\partial K_e}{\partial t}$	0.28	0.12	3.18	-1.13	1.58	2.71	61.50
RGz	-2.17	-1.00	7.60	-5.22	1.56	6.78	220.06
RKz	12.56	8.69	40.49	-8.68	32.71	41.39	561.24
RGe	2.10	1.20	5.15	-0.40	4.21	4.61	97.25
RKe	-7.59	-4.63	10.62	-11.48	-1.04	10.43	138.42

materially: in the correction-only paired control, the pooled median C_A increases from 0.50 to $1.54 W m^{-2}$, while the pooled median C_K shifts from -1.91 to $-0.03 W m^{-2}$. Thus, the corrected climatology shows a substantially stronger $A_Z \rightarrow A_E$ conversion and a barotropic conversion whose typical value is close to neutral, with both $K_Z \rightarrow K_E$ and $K_E \rightarrow K_Z$ cases. The published characterization of a predominantly strong $K_Z \rightarrow K_E$ pathway is not supported by the corrected calculation.

For the lateral boundary terms, BA_E , BK_Z , BA_Z , and BK_E retain their published distributions because these terms are numerically unchanged by the toolkit correction. The boundary pressure-work terms do not. In the corrected phase statistics, mean $B\Phi_Z$ values are approximately 8.4 – $10.2 W m^{-2}$ and mean $B\Phi_E$ values are approximately -3.0 to $-1.0 W m^{-2}$, far smaller than the published values. The corrected calculation therefore does not support the published description of anomalously large pressure-work terms.

Overall, both directly computed generation terms (G_Z and G_E) are numerically unchanged by the toolkit correction. The residual terms RG_Z and RG_E , however, inherit the C_A correction through budget closure. Their differences from the direct generation terms cannot be assigned uniquely to diabatic heating, meridional heat

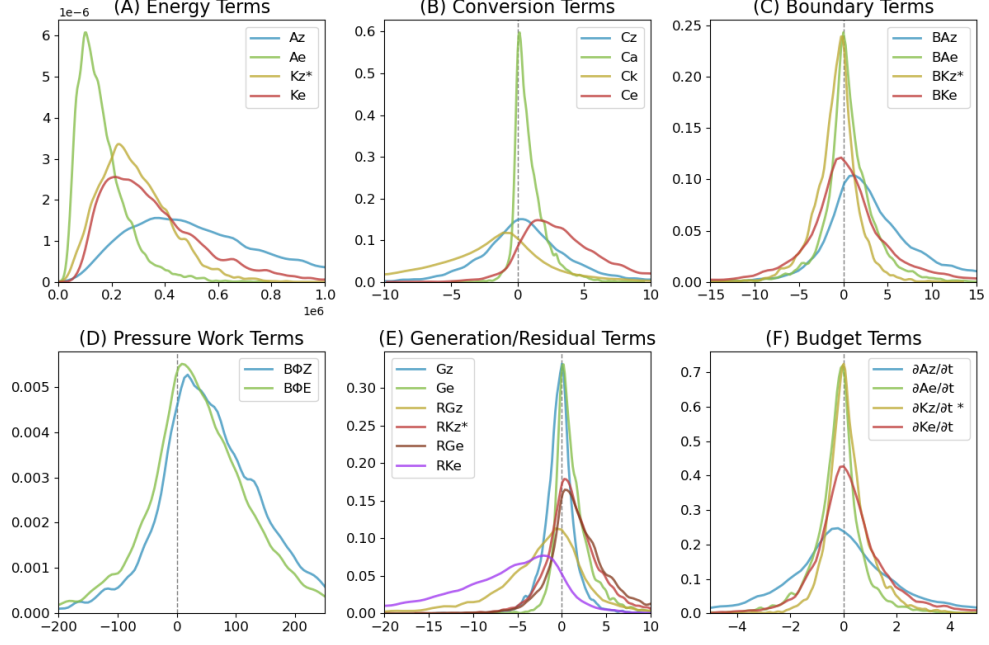


Fig. 3 Probability density function (PDF) plots for distinct Lorenz Energetics terms, computed as averages across the entire cyclone lifecycle: (A) energy, (B) conversion, (C) boundary, (D) pressure work, (E) generation and residual, and (F) budget terms. The x-axis represents the energy values, while the y-axis represents density counts. The units for the energy terms (A_Z , A_E , K_Z , and K_E) are in 10^5 J m^{-2} , and in W m^{-2} for the remaining terms. Terms related to K_Z are divided by 10 (*) to account for their order of magnitude difference, which would otherwise distort the visualization.

transport, or convective activity, because each residual also contains accumulated numerical error and unresolved contributions.

The composite kinetic residuals RK_Z and RK_E also inherit the C_K correction through budget closure. Their signs describe the net closure contribution but cannot be interpreted as frictional dissipation alone. In particular, the former explanation based on anomalously large $B\Phi_Z$ is no longer supported after the pressure-work correction.

3.2 Mean Energy Values Across Life Cycle Phases

The use of the CycloPhaser program allowed for the application of an objective criterion to dissect the cyclones into distinct life cycle phases, enabling an investigation of their energy cycles across these phases (Figure 4). Our analysis revealed that the mean energy flow directions rarely change across phases, although the magnitude of each term varies. However, the high standard deviation values indicate that energy fluxes frequently deviate from their mean direction.

The A_Z budget term exhibits mean positive values during the incipient phase, which turn negative during the intensification and mature phases, becoming positive

again during the decay phase. The mean K_Z budget values are positive throughout all phases except during the mature phase. In contrast, the A_E budget remains negative across the entire lifecycle, while the K_E budget is positive during the incipient and intensification phases but turns negative during the mature and decay phases. However, it is important to note that the large variability observed in these terms indicates that numerous individual cases deviate significantly from these mean behaviors, including instances of opposite signs or substantially greater magnitudes.

Across the different cyclone phases, the corrected mean behavior is dominated by baroclinic conversion. Mean C_A increases from 1.92 W m^{-2} in the incipient phase to 4.12 W m^{-2} during intensification, then decreases to 3.67 and 2.16 W m^{-2} during mature and decay phases. Mean C_K is 0.33 W m^{-2} during incipience, close to zero during intensification (-0.30 W m^{-2}), and modestly eddy-feeding during maturity and decay (-2.43 and -2.10 W m^{-2}). Barotropic conversion is therefore case-dependent and is not a continuous dominant source for eddies. The unchanged G_E term remains negative during incipience and positive thereafter, peaking during intensification, while G_Z remains negative throughout the life cycle.

Throughout the life cycle, the mean values indicate imports of A_Z , which decrease in the mature phase but rise again during the decay phase. The mean BA_E values exhibit a bimodal behavior, being positive during the incipient and intensification phases but turning negative during the mature and decay phases, though with negligible mean values in the latter. However, the large variability observed suggests that for individual cyclones, imports and exports of A_Z and A_E can differ substantially from these mean patterns, with some systems potentially exhibiting more pronounced or negligible boundary flux contributions.

Across all phases, RK_Z remains a large and highly variable composite closure term, while mean RK_E remains negative. The corrected pressure-work calculation removes the basis for attributing the magnitude of RK_Z to anomalously large $B\Phi_Z$, and neither residual should be read as a direct dissipation estimate.

3.3 Empirical Orthogonal Function Analysis

To better characterize the substantial variability observed in the LEC terms, we performed an Empirical Orthogonal Function (EOF) analysis to identify distinct energetic configurations throughout the cyclone lifecycle. In the corrected population, the leading mode explains 27.7% of the variance. The modes pattern-matched to published EOF2, EOF3, and EOF4 explain 12.0%, 10.5%, and 7.6%, respectively; the matched EOF4 is ranked fifth in the corrected solution, demonstrating that higher modes can exchange order. The first eight corrected modes explain 81.3% of the total variance. Comparisons below must therefore follow the one-to-one matched loading patterns rather than numerical rank alone.

Each cyclone can be described as a linear combination of the EOFs, with the corresponding principal components (PCs) indicating how strongly each mode contributes to the system's energetic structure. Consequently, the importance of a given EOF varies from system to system. The energy fluxes associated with each EOF represent deviations from the climatological mean (Figure 4), either reinforcing or weakening

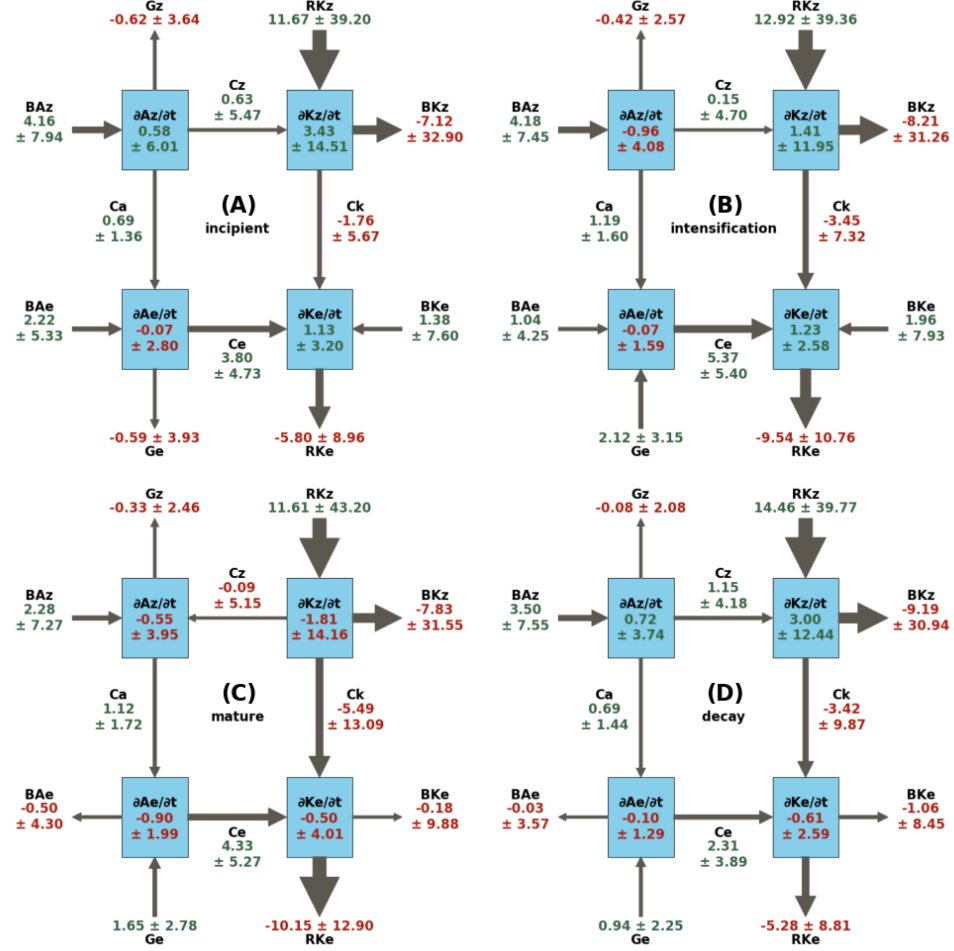


Fig. 4 Mean values and standard deviation of the limited-area Lorenz Energy Cycle (LEC) terms for each life cycle phase : incipient (A), intensification (B), mature (C), and decay (D). Each box represents an energy component ($J m^{-2}$), with arrows showing the direction of energy flow ($W m^{-2}$). The numbers adjacent to the arrows indicate the magnitude and direction of the energy flux, with green indicating positive values and red indicating negative values

specific energy pathways. In this way, each EOF reflects the amplification or attenuation of a characteristic energy transfer pattern within the LEC. In section 3.4, we examine cyclones that project strongly onto each EOF individually.

Figure 5 illustrates the Lorenz Energy Cycle (LEC) terms associated with EOF 1. Across all life cycle phases, there is a consistent enhancement of moist baroclinic conversions, characterized by the positive values of the $A_Z \rightarrow A_E \rightarrow K_E$ conversions, in conjunction with an enhancement of the G_E term. Furthermore, there is an increase in the $K_Z \rightarrow K_E$ conversion (C_K becomes more negative), particularly during

the mature and decay phases. These variations largely reinforce the mean behavior observed across the cyclone's life cycle. In addition, A_Z imports are enhanced, while a negative tendency is observed in its budget and generation terms. Conversely, the K_Z boundary, budget, and residual terms show a positive tendency. Lastly, the C_Z term presents a weak positive signal during the incipient and decay phases, and a negative tendency during the intensification and mature phases.

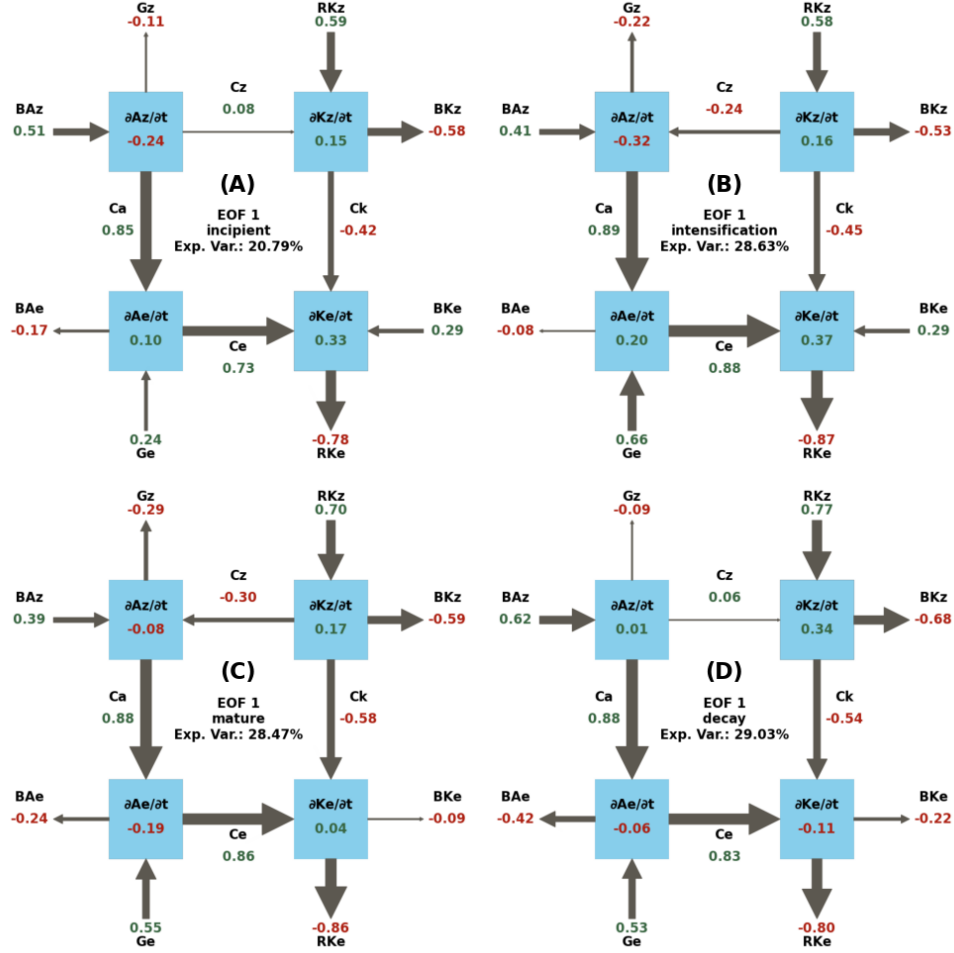


Fig. 5 Lorenz Energy Cycle (LEC) terms for the first Empirical Orthogonal Function (EOF 1) across different life cycle phases: (A) incipient, (B) intensification, (C) mature, and (D) decay.

During the incipient (Figure 6A) and intensification phases (Figure 6B), EOF 2 displays an overall weakening of the C_A term, in contrast to EOF 1. Additionally, there is a reduction in barotropic conversions, indicating a $K_E \rightarrow K_Z$ tendency, alongside a

434 weakening of K_E imports. This reduction is compensated by an enhancement in A_E
 435 imports, which appears to be the primary mechanism driving the increase in the C_E
 436 term. In the mature phase (Figure 6C), the the signal for A_E imports reverses in sign,
 437 and an enhancement of the moist baroclinic chain occurs, accompanied by a slight
 438 increase in $K_Z \rightarrow K_E$ conversions and a more pronounced increase in K_E imports.
 439 During the decay phase (Figure 6D), both the moist baroclinic chain and K_Z imports
 440 show a negative tendency. However, the enhanced barotropic conversions, coupled
 441 with a weakening of dissipation (positive tendency for RK_E), result in a slight yet
 442 positive tendency in K_E , suggesting a slower dissipation process.

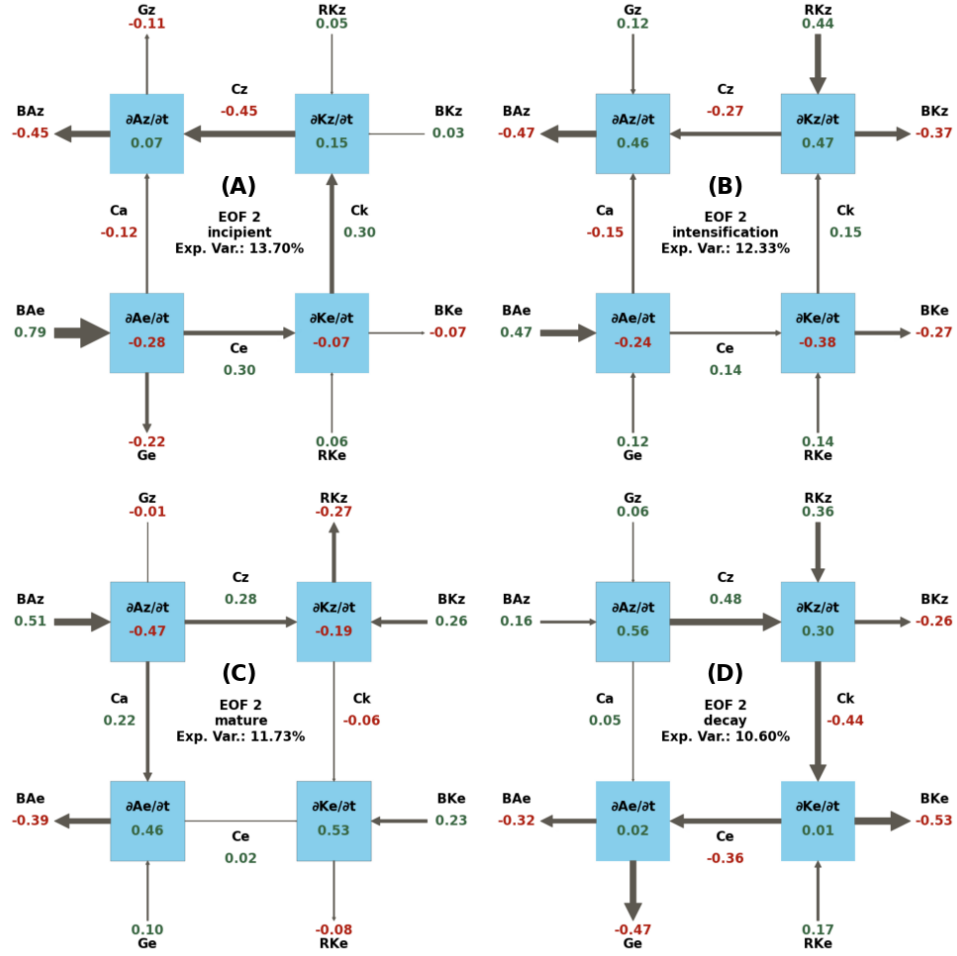


Fig. 6 Lorenz Energy Cycle (LEC) terms for the second Empirical Orthogonal Function (EOF 2) across different life cycle phases: (A) incipient, (B) intensification, (C) mature, and (D) decay.

For EOF 3 (Figure 7), the role of the moist baroclinic chain diminishes in eddy development. This is evident from the weakening of the C_E and G_E terms across the incipient (Figure 7A), intensification (Figure 7B), and mature (Figure 7C) phases, although the C_A term shows a slight enhancement during these phases. As a result, we can expect a weaker development for the incipient phase, indicated by the reduced K_E budget. However, during the intensification and mature phases, the dominant source of energy shifts to barotropic conversions. During the decay phase (Figure 7D), the marginal enhancement of the C_E term is driven by a slight increase in the G_E term and, more importantly, by an enhancement of A_E imports. Although this works in conjunction with increased K_E imports, the simultaneous enhancement of $K_E \rightarrow K_Z$ conversions leads to the dissipation of eddies.

During the incipient phase (Figure 8A) of EOF 4, the moist baroclinic chain weakens, as do the K_E imports, while the barotropic conversions are enhanced. As the system progresses to the intensification phase (Figure 8B), the barotropic conversion signal becomes neutral, and the weakening of the baroclinic chain lessens, accompanied by a strengthening of A_E imports. In the mature phase (Figure 8C), both the baroclinic chain and the A_E and K_E imports are slightly enhanced. However, the barotropic conversions reverse, signaling $K_E \rightarrow K_Z$ conversions, indicating a transfer of energy from eddy kinetic energy to zonal kinetic energy. In the decay phase (Figure 8D), the signal of baroclinic conversions returns to its initial weakened state and A_E imports are reduced, while K_E are enhanced. Although the $K_E \rightarrow K_Z$ conversions are enhanced, they are counterbalanced by the K_E imports, leading to a slower eddy dissipation (higher tendency for K_E the budget).

Here we present a summary of the energy pathways that were enhanced for each EOF, indicating the increased contribution to cyclone development (i.e., K_E increases). During EOF1, both the moist baroclinic and barotropic chains are enhanced, with increased imports of K_E during the incipient and intensification phases. For EOF2, cyclone development is characterized by the enhancement of the $BA_E \rightarrow A_E \rightarrow K_E$ chain during the incipient and intensification phases, followed by an increase in K_E imports during the mature phase. For EOF3, during the incipient phase, there is an enhancement of K_E imports, which is followed by an increase in barotropic conversions during the intensification and mature phases. For EOF4, barotropic conversions are enhanced during the incipient phase, with nearly neutral baroclinic and barotropic conversions during intensification, and an enhanced baroclinic chain along with K_E imports during the mature phase. During cyclone decay, EOF1 shows an enhancement in dissipation, EOF2 and EOF4 present an enhancement in exports of K_E , and EOF3 exhibits an increase in $K_E \rightarrow K_Z$ conversions.

3.4 Characteristics of Cyclones Based on EOF Signal

In this section, as the characteristics of the systems associated with each EOF are analyzed, the EOF signal becomes relevant, necessitating a distinction between systems related to positive EOF signals — EOF(+) — and negative EOF signals — EOF(-), respectively. EOF1 accounts for the largest number of cyclones among the first four EOFs for both EOF(+) and EOF(-). For EOF(+), it is followed by EOF2, EOF3, and EOF4, while for EOF(-), it is followed by EOF3, EOF2, and EOF4, respectively.

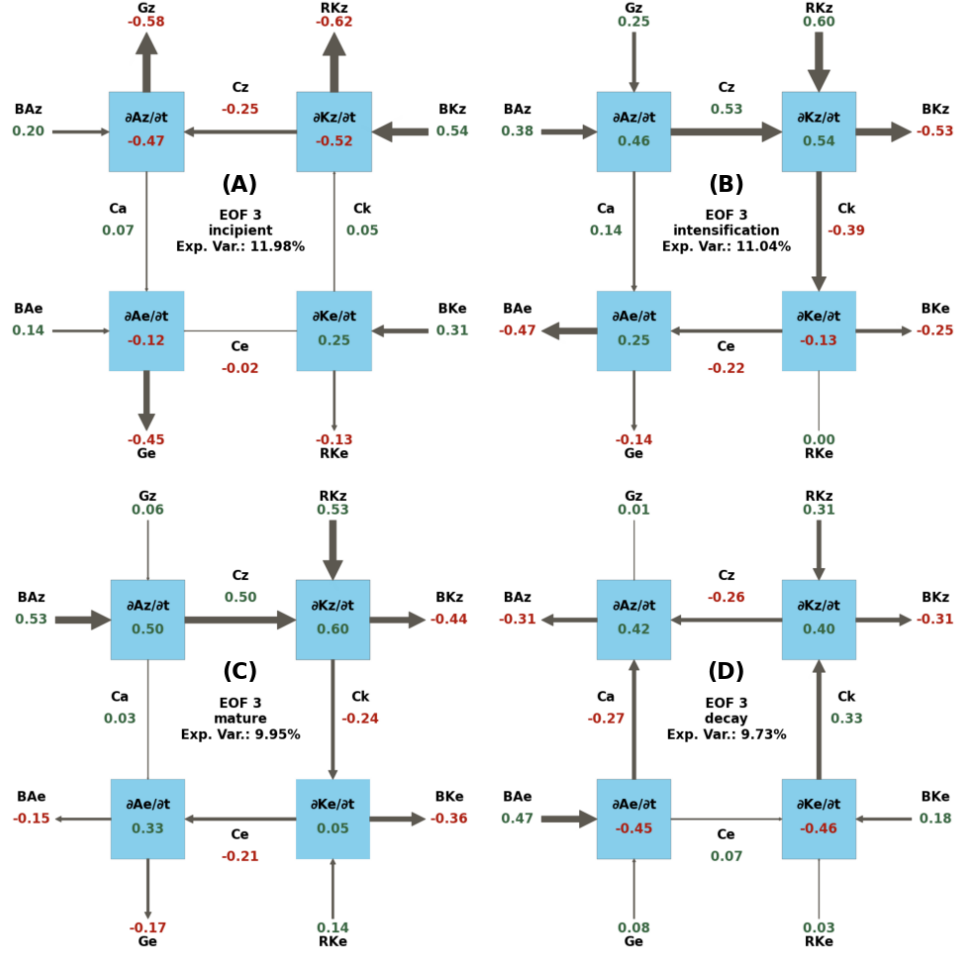


Fig. 7 Lorenz Energy Cycle (LEC) terms for the third Empirical Orthogonal Function (EOF 3) across different life cycle phases: (A) incipient, (B) intensification, (C) mature, and (D) decay.

(Figure S1a). The track densities for the cyclones associated with EOF(+) are presented in Figure 9, while those for EOF(-) are shown in Figure 10. For EOF1(+), the track density maximum extends from SE-BR and LA-PLATA toward the ARG region, highlighting the significant contribution of systems originating in the former regions compared to the latter, as also indicated in Figure 11a. In contrast, for the remaining EOFs(+), the track density maximum is located near the ARG region, as most systems originate there. An exception is observed in EOF3(+), where the track density maximum extends northeastward due to the increased contribution of SE-BR systems to this EOF.

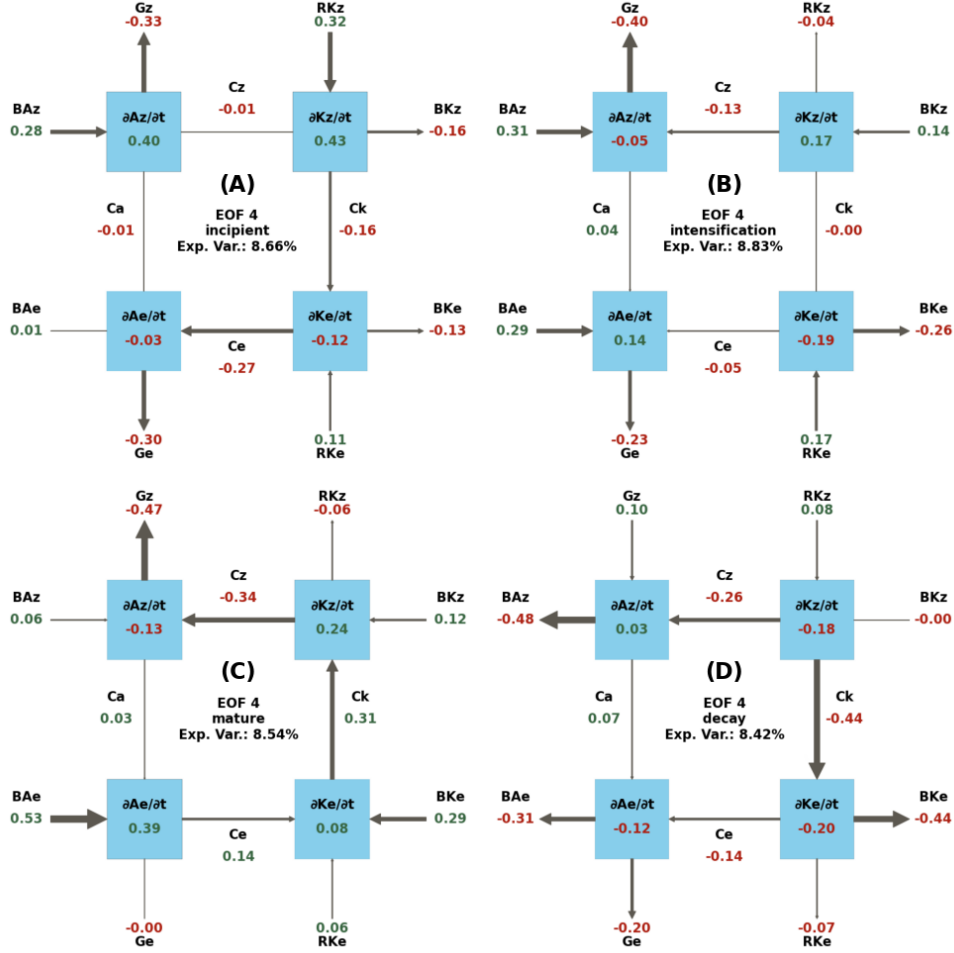


Fig. 8 Lorenz Energy Cycle (LEC) terms for the fourth Empirical Orthogonal Function (EOF 4) across different life cycle phases: (A) incipient, (B) intensification, (C) mature, and (D) decay.

Meanwhile, EOFs(-) exhibit a more variable behavior. EOF1(-) displays two track density maxima: one near SE-BR and another near ARG, as fewer systems originate from LA-PLATA for this EOF (Figure 11c). EOF2(-) is predominantly influenced by ARG systems, whereas EOF3(-) shows an increased contribution from LA-PLATA systems, with a primary track density maximum near ARG and a secondary maximum over the La Plata River mouth. Lastly, EOF4(-) presents a track density maximum between LA-PLATA and SE-BR, reflecting the increased relative contribution of systems from these regions to this EOF.

Despite regional differences in track density maxima, nearly all EOFs exhibit a common pattern of cyclone tracks extending southeastward across the South Atlantic

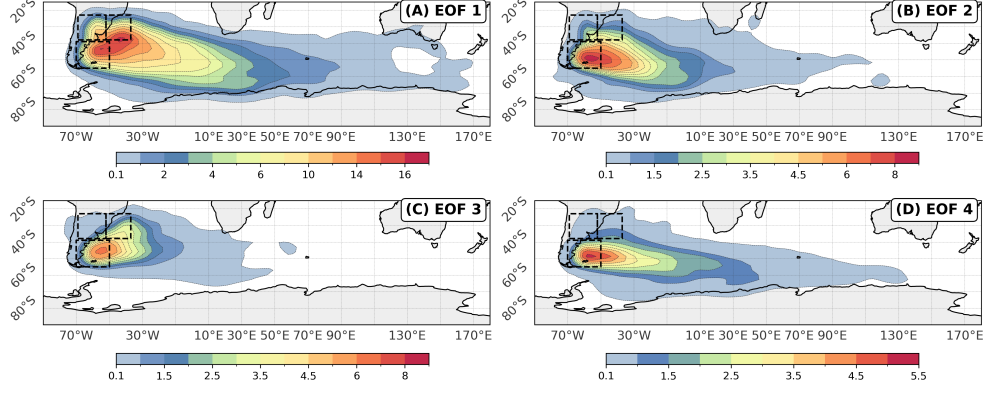


Fig. 9 Cyclone track density for the systems associated with EOFs(+): (A) EOF1, (B) EOF2, (C) EOF3, and (D) EOF4. The track density unit is cyclones per 10^6 km^2 per month. The dashed rectangles mark the genesis regions shown in Figure 1.

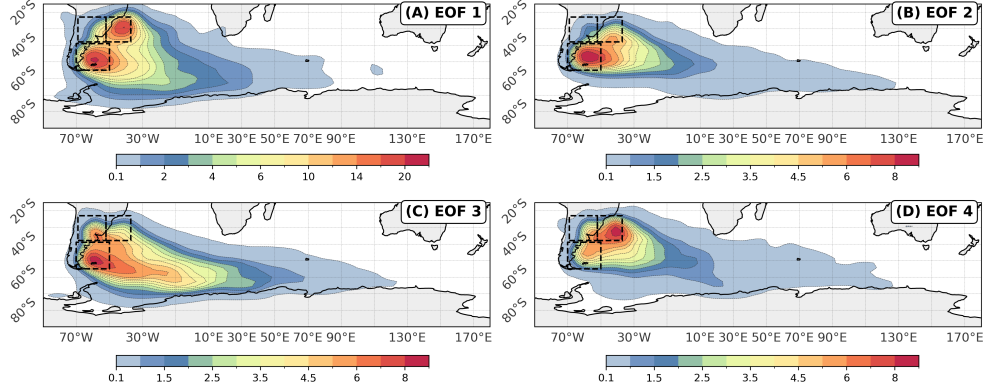


Fig. 10 Cyclone track density for the systems associated to EOFs(-): (A) EOF1, (B) EOF2, (C) EOF3 and (D) EOF4. The track density unit is cyclonic centers per 10^6 km^2 per month. The track density unit is cyclones per 10^6 km^2 per month. The dashed rectangles mark the genesis regions shown in Figure 1.

(Figures 9 and 10), with the exception of EOF3(+). Some systems display exceptional mobility, reaching as far as $150^\circ E$. This behavior aligns with the typical lifecycle of cyclonic systems originating near the South American coast, where cyclones generally develop near the coastal region, mature as they propagate southeastward, and decay closer to the Antarctic region (de Souza et al., 2024).

As shown in Figure 11b, EOF1(+) and EOF4(+) systems predominantly originate during JJA, while EOF2(+) and EOF3(+) genesis events are more evenly distributed across the seasons. For EOFs(-), EOF1(-) systems occur mainly during DJF, whereas EOF4(-) systems are more frequent during DJF and MAM. In contrast,

515 EOF2(-) and EOF3(-) systems are predominantly observed during JJA and SON
 516 (Figure 11d). The seasonal distribution of EOF2(+) and EOF3(+) systems aligns
 517 with the expected behavior, as ARG systems are well-distributed throughout the year
 518 (Gramscianinov et al., 2019; Crespo et al., 2021). However, for the remaining EOFs,
 519 the observed behavior suggests that seasonal variations are linked to changes in the
 520 systems' energetics.

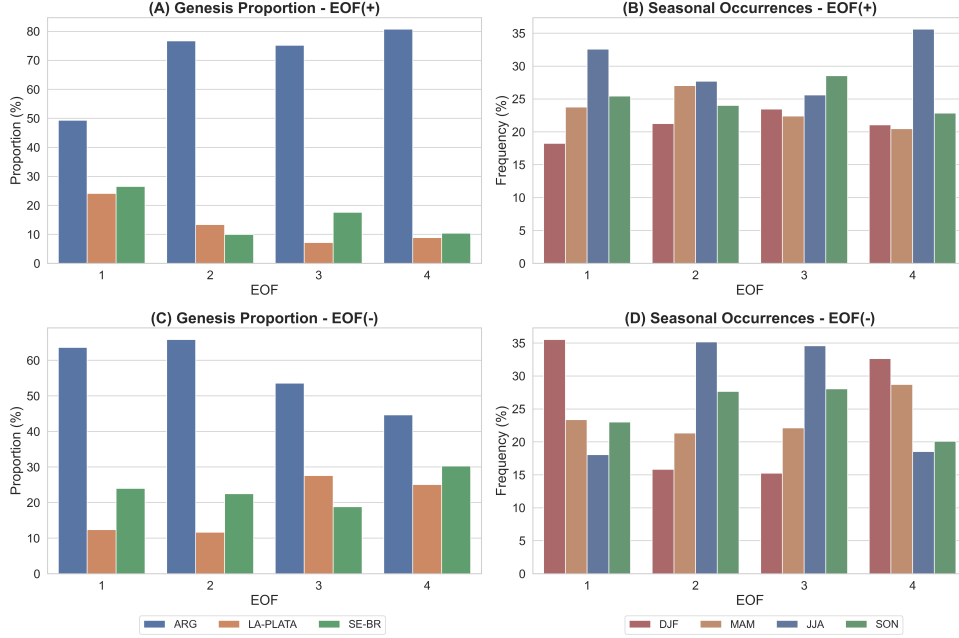


Fig. 11 Genesis proportion and seasonal occurrences of EOFs. (A) Genesis proportion for EOF(+), (B) Seasonal occurrences for EOF(+), (C) Genesis proportion for EOF(-), and (D) Seasonal occurrences for EOF(-). The colors represent different genesis regions (ARG, LA-PLATA, SE-BR) in (A) and (C), while in (B) and (D), the colors correspond to seasonal distributions (DJF, MAM, JJA, SON).

521 3.5 Groups of Intense Systems

522 While the previous sections examined the LEC for all cyclonic systems in the South
 523 Atlantic — analyzing both the mean behavior and its associated variability through
 524 EOF analysis — this section focuses on the energetic groups associated with the most
 525 intense cyclones in the dataset. The K-Means algorithm was applied to the first eight
 526 PCs of the selected cyclones, identifying five distinct LEC groups, as determined by
 527 the Elbow Method. Following the cluster identification analysis, the energetic char-
 528 acteristics of the systems were reconstructed, with the mean behavior for each group
 529 represented in Figure 12.

530 The canonical corrected reconstruction likewise uses five groups, fitted independently to the first eight aligned total-lifecycle PC scores. It includes 1,486 corrected
531 intense cyclones, compared with 1,744 in the archived analysis, and matches corrected
532 group centroids to the legacy centroids for comparison. Cluster labels facilitate
533 comparison but do not imply identical physical membership.
534

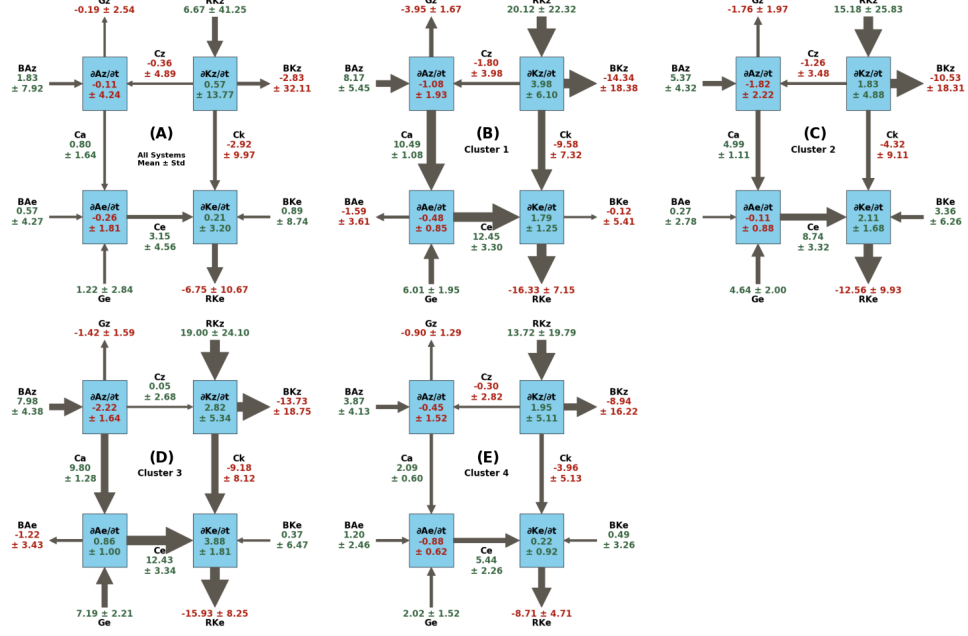


Fig. 12 (A) Mean values and standard deviations of the limited-area Lorenz Energy Cycle (LEC) terms for all analyzed systems. Panels (B–F) illustrate the reconstructed LEC for the five clusters identified among the most intense cyclones in the dataset, defined as systems with maximum central relative vorticity at 850 hPa exceeding the 90th percentile.

535 Comparing the energy fluxes of the clusters with the climatological mean computed across all cyclones provides insight into the mechanisms that distinguish these
536 systems as the most intense (Figure 12). The corrected reconstruction retains five energetically distinct groups, but the published cluster-by-cluster pathway descriptions cannot be carried over without revision because both affected conversion terms
537 enter the clustering and group reconstruction. The composite residuals also cannot be
538 interpreted as direct dissipation estimates.
539

540 Figure 13 displays the track density for all clusters. For all clusters, the track density maxima extend from a region near the southeastern South American coast, between 30°S and 50°S, propagating southeastward toward the Antarctic region. Cluster 2 and 3 present tracks more concentrated near the continent, related to their lower mean duration (Figure S2). In contrast, Cluster 1 presents a track maxima spread
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543
544
545
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from near $60^{\circ}E$ to near $10^{\circ}E$. Meanwhile, Cluster 4 presents a maxima near the continent and other near Antarctica. While the track density maxima near LA-PLATA and SE-BR are associated with the incipient and intensification phases of these systems, the maxima near Antarctica correspond to their decay phase (de Souza et al., 2024). The relatively small displacement of these systems indicate that they might be related to explosive cyclogenesis and in fact, among explosive cyclogenesis in South America, evidence suggests a great contribution from LA-PLATA (Andrade et al., 2024, 2025), however, further investigation is needed to confirm this.

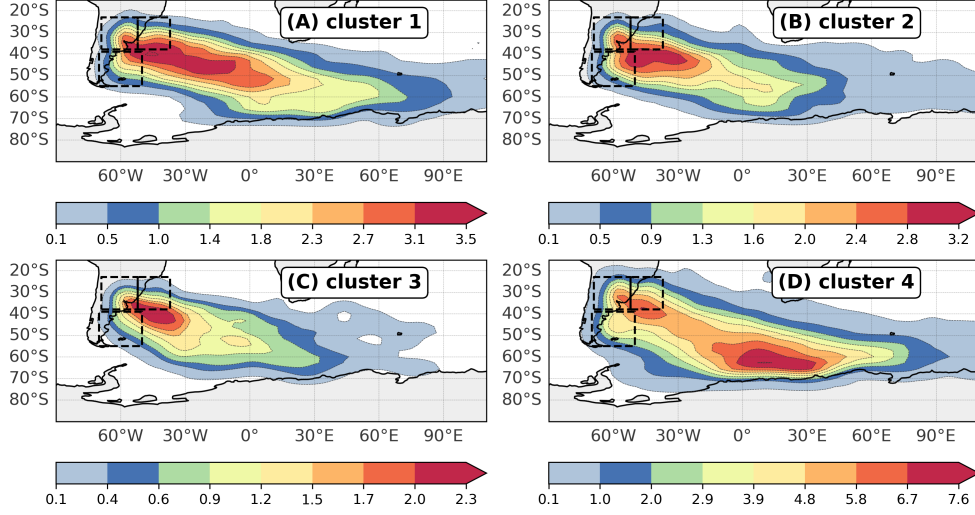


Fig. 13 Cyclone track density for the five clusters (A–E) identified among the most intense cyclones in the dataset. These systems are defined as those with maximum central relative vorticity at 850 hPa exceeding the 90th percentile. The dashed rectangles mark the genesis regions shown in Figure 1.

In the corrected five-group reconstruction, the matched groups contain 647, 266, 229, 206, and 138 systems. Across the full corrected intense subset, 53.3% originate in ARG, 28.7% in LA-PLATA, and 18.0% in SE-BR; JJA is the most frequent season (27.4%), closely followed by SON (26.1%). LA-PLATA remains overrepresented relative to its 18.9% share of the full published dataset, but it is not the most frequent absolute genesis region. Group-specific labels and interpretations require care because the corrected clusters are matched by centroid and do not preserve membership.

4 Discussion

4.1 Key Features

The LEC diagrams (Figure 4) indicate that the mean energetic behavior is preserved across distinct life cycle phases. However, the high variability shown in Figure 3 reveals a wide range of behaviors among South Atlantic cyclones. While this section focuses

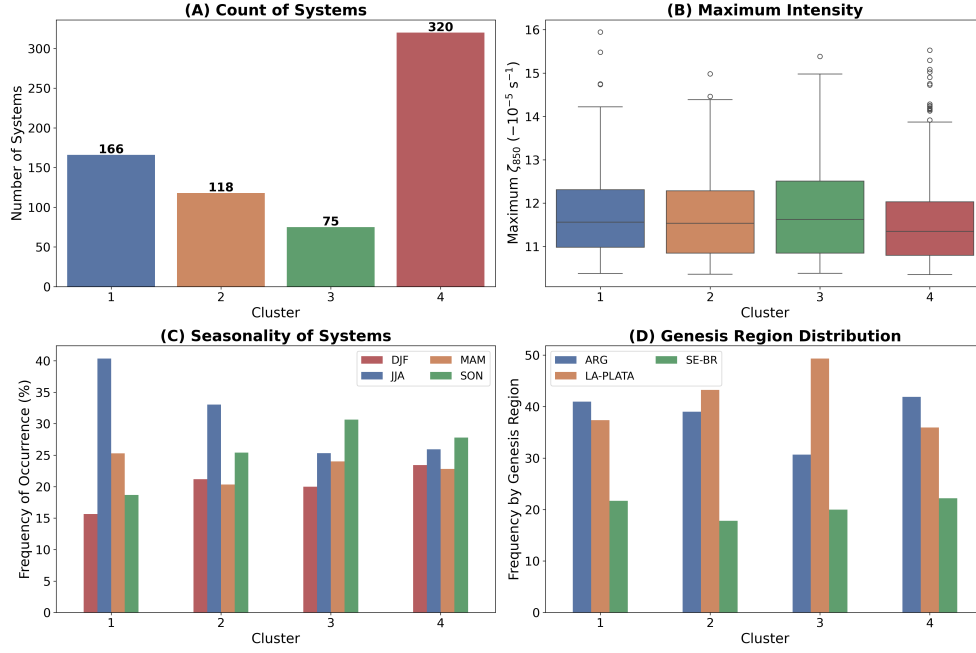


Fig. 14 Characteristics of the five clusters (A–E) identified among the most intense cyclones in the dataset. (A) Number of cyclones per cluster, (B) maximum intensity (defined as maximum central relative vorticity at 850 hPa), (C) seasonal distribution of occurrences, and (D) genesis region count. In (C), colors represent different seasons (DJF, MAM, JJA, SON), while in (D), colors correspond to genesis regions (ARG, LA-PLATA, SE-BR).

on the mean energy cycle, it is important to recognize that energy budget terms serve primarily as a starting point. The energy conversions and fluxes offer more robust insight into the dynamical processes governing cyclone development. In addition to the phase-specific diagrams, a schematic of the mean Lorenz Energy Cycle for all systems is shown in Figure 15, facilitating comparison across phases and highlighting dominant pathways.

The A_Z term increases during the early phases due to persistent influxes across the boundaries ($BA_Z > 0$), rather than local generation (G_Z). As baroclinic conversions intensify, the A_Z budget becomes negative during the intensification and mature phases, before returning to positive in the decay phase. This evolution suggests that A_Z variations are not driven by latitudinal diabatic heating contrasts. A similar pattern was reported by Dias Pinto and Rocha (2011), although their methodology excluded the incipient phase due to the use of sea-level pressure for cyclone tracking, instead of relative vorticity at 850 hPa (Sinclair, 1994; Hoskins and Hodges, 2002).

For K_Z , a general increase is observed except during the mature phase. The corrected mature-phase mean C_K remains negative, but its magnitude is substantially reduced relative to the published value and is smaller than the corrected C_A . Notably, BK_Z and RK_Z present high magnitudes: the former indicates export of zonal kinetic

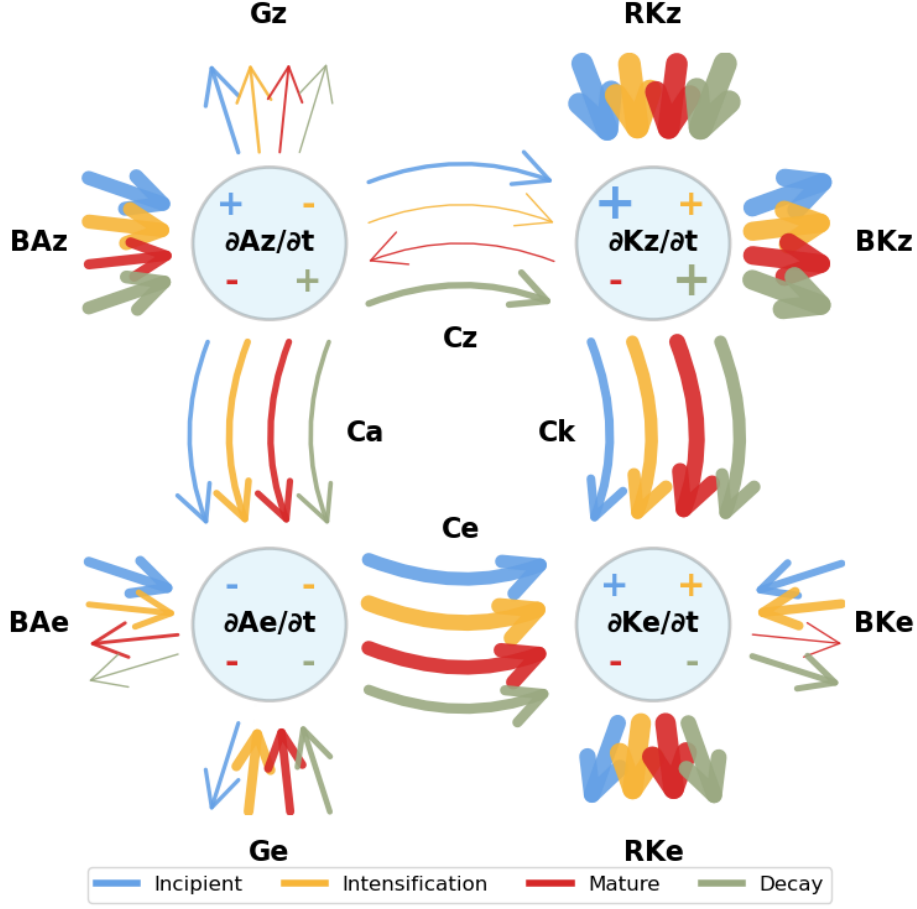


Fig. 15 Schematic representation of the mean Lorenz Energy Cycle (LEC) for all analyzed cyclones in the South Atlantic, combining information from all life cycle phases. Arrows represent energy fluxes between reservoirs, with colors indicating the life cycle phases. The arrow thicknesses and symbol sizes for the balance terms ($\partial/\partial t$) are proportional to the absolute magnitude of each term, as presented in Figure 4.

energy, while the latter is a composite closure term that aggregates signed frictional dissipation, boundary pressure work ($B\Phi_Z$), unresolved-scale transfers, and numerical errors.

As the Semi-Lagrangian framework does not account for changes in the background energetics along a moving domain, part of the apparent time evolution in A_Z and K_Z may simply reflect the cyclone's poleward drift (de Souza et al., 2024). Such drift carries the system into latitudes of larger climatological background APE, implying an increase in A_Z (Novak and Tailleux, 2018; Liu et al., 2024). Because the zonally

integrated K_Z reaches its climatological maximum near 60°S (Novak and Tailleux, 2018; Liu et al., 2024), K_Z should decrease for storms that originate over ARG and migrate poleward, but increase for those formed in SE-BR or LA-PLATA. In practice, however, both reservoirs evolve irregularly and non-monotonically, indicating that intrinsic cyclone dynamics, rather than passive domain advection, control the local energetics, corroborating Federer et al. (2025).

The A_E budget is negative throughout the life cycle, with minimal values during the incipient and intensification phases. This reflects a highly active and efficient baroclinic chain. Energy is transferred from A_Z via C_A , and from A_E to K_E via C_E , driven by meridional heat transport and vertical motions in frontal zones. During intensification, G_E becomes positive, associated with convective heating along the cold front (e.g., Govekar et al., 2011). Simultaneously, radiative heating in the warm sector due to long-wave absorption from mid- to high-level clouds (Lau and Crane, 1997; Keshtgar et al., 2023), combined with latent heat release from intense precipitation during this phase (McErlich et al., 2023), enhances G_E . As the system matures and convection weakens, BA_E reverses (export), and the A_E budget sharply decreases. During decay, with disorganized frontal structures and weak diabatic forcing, A_E returns to near-zero budget values.

The K_E reservoir increases during incipient and intensification phases, when baroclinic conversion and boundary imports supply energy. Corrected C_K is weak and of mixed sign in these early phases. During the mature and decay phases, mean C_K is eddy-feeding but remains smaller in magnitude than C_A , while changes in boundary fluxes and the composite RK_E accompany net energy loss. Because RK_E includes several unresolved contributions, it does not by itself demonstrate dominant physical dissipation.

Both $B\Phi_Z$ and $B\Phi_E$ were recomputed with corrected boundary formulations. Their magnitudes collapse relative to the published values, and $B\Phi_E$ changes sign in most paired cyclone phases. The corrected results therefore remove the basis for attributing their published amplitudes to ERA5 geopotential errors or for describing those amplitudes as a physical feature of the cyclones.

4.2 Cyclone Groups

Substantial variability in the dataset, as revealed by the EOF analysis, is largely captured by EOF1(+) and EOF1(-), which together represent nearly 40% of all systems. EOF1(+) is linked to stronger cyclones, while EOF1(-) corresponds to weaker ones (Supplementary Figure S1b), indicating a bi-modal structure in cyclone intensity. This suggests that "day-to-day" systems tend to follow distinct energetic regimes. Traditional cyclone climatologies over the South Atlantic tend to group all cyclones together (Sinclair, 1994; Hoskins and Hodges, 2005; Reboita et al., 2010; Gramscianinov et al., 2019, e.g.). In contrast, the results presented here suggest that these systems exhibit distinct behaviors and may therefore be categorized into separate groups.

These intensity differences manifest in their energy cycles and physical characteristics. Figure 16 presents schematic LECs for the four leading EOF(+) groups. EOF1(+) displays a consistent enhancement across all energy pathways, while EOF1(-) shows a reduction. These differences extend to cyclone mobility: EOF1(+) systems follow

the South Atlantic storm track (Hoskins and Hodges, 2005; Gramscianinov et al., 2019), whereas EOF1(-) systems remain more confined near the South American coast (Figure 10a), particularly within the ARG and SE-BR regions (Figure 11d), and exhibit lower translational speeds (Figure S1).

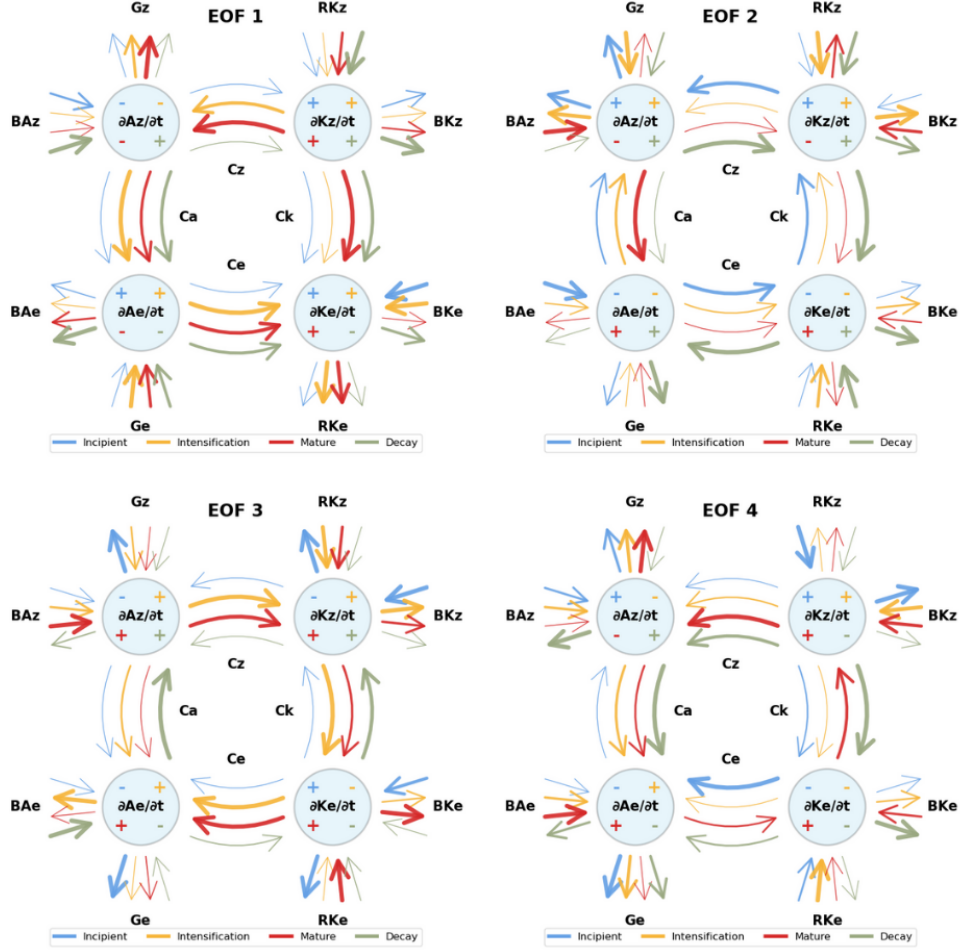


Fig. 16 Schematic representation of the mean Lorenz Energy Cycle (LEC) for cyclones associated with the four leading positive Empirical Orthogonal Functions (EOF1(+) to EOF4(+)). Arrows represent energy fluxes between reservoirs, with colors indicating life cycle phases. The thickness of each arrow and the size of the balance terms ($\partial/\partial t$) symbols are proportional to the absolute magnitude of each term.

EOF2–EOF4 describe less frequent cyclone types. EOF2(+), EOF3(-), and EOF4(-) are associated with relatively stronger systems, while their counterparts correspond to weaker ones (Figure S1b). EOF2(+) is notable for lacking enhancements

in baroclinic/barotropic conversions or K_E imports during early phases. EOF3(-) and EOF4(-), in contrast, display a pronounced intensification of the $G_E \rightarrow A_E \rightarrow K_E$ chain, especially during the incipient and intensification phases. These patterns indicate that despite a shared dominant energy structure, South Atlantic cyclones express diverse configurations in their development. This variability is further reflected in differences in genesis region and seasonality (Figure 11), with EOF1(+) and EOF3(-) dominating among the most intense systems (Supplementary Figure S3).

For corrected intense cyclones (above the 90th percentile in central vorticity), genesis is most frequent in ARG (53.3%), followed by LA-PLATA (28.7%) and SE-BR (18.0%). JJA is the most frequent season (27.4%), closely followed by SON (26.1%). These systems retain a strong moist-baroclinic pathway, while the corrected cluster memberships and pathway proportions require the replacement cluster figures for final interpretation.

The corrected reconstruction contains five, not four, LEC-based clusters. In the canonical matched numbering, Group 1 contains the largest number of corrected systems (647) and the lowest mean maximum central vorticity ($9.84 \times 10^{-5} s^{-1}$). The five-group solution continues to demonstrate diversity among intense cyclones, but changes in membership and in the affected conversion terms preclude carrying over the published cluster-specific physical narratives unchanged.

4.3 Mechanisms of Energy Transfer

The corrected results identify baroclinic conversion (C_A), rather than barotropic conversion (C_K), as the dominant conversion mechanism for cyclone development in the South Atlantic. In the correction-only paired control, median C_A increases from 0.50 to $1.54 W m^{-2}$, whereas median C_K shifts from -1.91 to $-0.03 W m^{-2}$. The proportion of cyclone phases in which an eddy-feeding C_K exceeds C_A falls from 64.7% to 29.3%, and to 19.2% during intensification. Barotropic conversion remains an eddy-energy source in a substantial subset of cases, particularly later in the lifecycle, but it is neither ubiquitous nor typically two to three times larger than baroclinic conversion.

Despite LEC-based studies dating back decades, few have examined South Atlantic cyclones, limiting direct comparisons. Case studies on tropical systems emphasize the $G_E \rightarrow A_E \rightarrow K_E$ chain and barotropic conversions (Brennan and Vincent, 1980; Veiga et al., 2008), while subtropical systems show mixed roles for both baroclinic and barotropic conversions (Michaelides, 1987; Dias Pinto et al., 2013; Pezza et al., 2014; Cavicchia et al., 2018). For extratropical systems, energy pathways vary: some are baroclinically dominated (Dias Pinto and Rocha, 2011; Black and Pezza, 2013), others barotropically (Michaelides, 1992; Dias Pinto and Rocha, 2011), and some combine both with varying degrees of K_E import (Wahab et al., 2002; Bulic, 2006; Pezza et al., 2010). The corrected climatology supports a dominant baroclinic pathway with a variable, secondary barotropic contribution; it does not support a consolidated picture of barotropic dominance.

It is also important to emphasize that, while previous studies investigated the LEC using an Eulerian framework, this study employed a Semi-Lagrangian framework. This approach has the advantage of isolating the energetics strictly related to the

target system. However, the extent to which this methodology differs from the Eulerian framework in representing the magnitude and direction of energy fluxes has not been extensively assessed, apart from one study by Michaelides et al. (1999). Also, the extent to which our findings generalize to other regions remains uncertain. The corrected results do not provide evidence that barotropic conversion is more pronounced in the South Atlantic than in the North Atlantic.

The corrected C_K evolution also changes the comparison with earlier decay-phase studies. In the paired control, phase medians are positive during incipience and intensification (0.53 and 0.10 W m^{-2}) and modestly negative during maturity and decay (-0.65 and -0.75 W m^{-2}). The published progression from strongly eddy-feeding conversion in maturity toward neutrality during decay is therefore not reproduced. Barotropic conversion can feed eddies during later phases, but the corrected statistics do not support the claim that it continuously sustains most systems through decay.

Some limitations of our approach must be acknowledged. Although the Semi-Lagrangian framework follows the cyclone’s trajectory, it does not account for regional variations in background APE and kinetic energy. This may influence A_Z , K_Z , and their boundary terms as the system moves through varying energy environments. Furthermore, the LEC formulation relies on a global reference state for APE. In contrast, the local APE framework proposed by Novak and Tailleux (2018) defines APE relative to each air parcel’s individual reference state. While this approach can yield additional insights—such as the potential for cyclones to locally generate APE (Federer et al., 2024), the overall global energy fluxes remain consistent between the global and local formulations (Liu et al., 2024). Moreover, the modest BA_Z values observed here (Figure 4) may be partially explained by APE transport from the polar upper troposphere, as shown for the North Atlantic storm tracks by Federer et al. (2025).

5 Summary and Conclusions

In this study, we examined the Lorenz Energy Cycle (LEC) for a validated corrected population of 3,820 of the 6,789 archived cyclonic systems originating in the South-western Atlantic. Comparisons with the full published population therefore combine the toolkit correction and a population change. A notable exception to this large-sample climatological approach is the work by Black and Pezza (2013), which focused solely on explosive cyclones and used large fixed domains, contrasting with the Semi-Lagrangian approach adopted in this study. While studies such as Smith (1980) provide valuable reviews of the energetics of extratropical cyclones, focusing on broad averages and generalized mechanisms, and case studies have explored the dynamics of specific systems (Pezza et al., 2014; Dias Pinto and Rocha, 2011; Cavicchia et al., 2018, e.g.), our study tracks the energy cycles of individual cyclones across distinct life cycle phases.

The corrected results demonstrate a clear baroclinic energy pathway for most cyclonic systems in the South Atlantic. Baroclinic conversion peaks during intensification and is the dominant conversion throughout the mean lifecycle. Barotropic conversion is close to neutral in the pooled median, changes sign across phases and

730 cases, and exceeds baroclinic conversion in only 29.3% of paired cyclone phases. It
 731 remains a relevant pathway for a subset of systems, but the published claim that it
 732 is typically two to three times larger than baroclinic conversion does not survive the
 733 correction. The unchanged generation of eddy potential energy peaks during intensi-
 734 fication, while eddy kinetic energy imports occur primarily during the incipient and
 735 intensification phases.

736 Despite the high variability among the systems, the leading corrected EOF remains
 737 recognizable, while higher modes can exchange rank and must be compared by
 738 matched loading pattern. This heterogeneity is also evident among the most intense
 739 systems, for which the corrected reconstruction retains five distinct groups. Across that
 740 corrected subset, most systems originate in ARG, although LA-PLATA is overrepre-
 741 sented relative to its share of the full published dataset. Final cluster-specific physical
 742 interpretations require the corrected-only replacement figures and author review.

743 The goal of this study was to investigate the energy cycle of cyclones in the South
 744 Atlantic region using a Semi-Lagrangian approach. Previous studies for this region
 745 have assessed the distinct synoptic and dynamical conditions related to cyclogenesis
 746 in the three primary genesis regions, as well as the seasonal variability in their devel-
 747 opment (Gramscianinov et al., 2019; Crespo et al., 2021). A distinction in the energy
 748 cycles among these genesis regions, along with their seasonal variability, remains valu-
 749 able. The corrected results motivate investigation of when and where the weaker,
 750 case-dependent barotropic pathway feeds eddy kinetic energy, rather than assuming
 751 that it is a ubiquitous dominant source near the cyclone center.

752 Appendix

753 Mathematical expressions used for the calculation of the Lorenz Energy Cycle (LEC),
 754 adopting the notation from Michaelides (1987).

755 Firstly, we define the zonal mean of a variable X , between longitudes λ_1 and λ_2 :

$$[X]_{\lambda} = \frac{1}{\lambda_2 - \lambda_1} \int_{\lambda_2}^{\lambda_1} X d\lambda \quad (13)$$

756 The eddy component of this variable is its deviation from the zonal mean:

$$(X)_{\lambda} = X - [X]_{\lambda} \quad (14)$$

757 The domain mean of the variable X , defined over the computational domain
 758 bounded by longitudes λ_1 and λ_2 , and latitudes ϕ_1 and ϕ_2 , is given by:

$$[X]_{\lambda\phi} = \left(\frac{1}{\lambda_2 - \lambda_1} \right) \left(\frac{1}{\sin \phi_2 - \sin \phi_1} \right) \int_{\lambda_2}^{\lambda_1} X \cos \phi d\lambda d\phi \quad (15)$$

759 Similarly, we define the deviation of the zonal mean from the domain mean:

$$([X]_{\lambda})_{\phi} = [X]_{\lambda} - [X]_{\lambda\phi} \quad (16)$$

760 From the definitions above, the four energy components used in the LEC
 761 computation are defined as follows:

$$A_Z = \int_{p_t}^{p_b} \frac{[(T)_\lambda]^2 \lambda_\phi}{2[\sigma]_\lambda \phi} dp \quad (17)$$

$$A_E = \int_{p_t}^{p_b} \frac{[(T)_\lambda]^2 \lambda_\phi}{2[\sigma]_\lambda \phi} dp \quad (18)$$

$$K_Z = \int_{p_t}^{p_b} \frac{[(u)_\lambda]^2 + [(v)_\lambda]^2 \lambda_\phi}{2g} dp \quad (19)$$

$$K_E = \int_{p_t}^{p_b} \frac{[(u)_\lambda]^2 + [(v)_\lambda]^2 \lambda_\phi}{2g} dp \quad (20)$$

where p is the atmospheric pressure, with subscripts b and t denoting the lower (base) and upper (top) pressure boundaries of the atmosphere, respectively. T represents temperature, g is the acceleration due to gravity, and u and v are the zonal and meridional wind components, respectively. The static stability parameter σ is defined as:

$$\sigma = \left[\frac{gT}{c_p} - \frac{pg}{R} \frac{\partial T}{\partial p} \right]_{\lambda\phi} \quad (21)$$

where c_p is the specific heat at constant pressure, and R is the ideal gas constant for dry air.

The four conversion terms are defined as follows, integrating over the atmospheric column from the base (p_b) to the top (p_t) pressures:

$$C_Z = \int_{p_t}^{p_b} -[(T)_\lambda]_\phi ([\omega]_\lambda)_\phi \lambda_\phi \frac{R}{gp} dp \quad (22)$$

$$C_E = \int_{p_t}^{p_b} -[(T)_\lambda (\omega)_\lambda]_\phi \lambda_\phi \frac{R}{gp} dp \quad (23)$$

$$C_A = \int_{p_t}^{p_b} - \left(\frac{1}{a\sigma} \left[(v)_\lambda (T)_\lambda \frac{\partial ([T]_\lambda)_\phi}{\partial \phi} \right]_{\lambda\phi} + \frac{1}{\sigma} \left[(\omega)_\lambda (T)_\lambda \frac{\partial ([T]_\lambda)_\phi}{\partial p} \right]_{\lambda\phi} \right) dp \quad (24)$$

$$C_K = \int_{p_t}^{p_b} \frac{1}{g} \left(\left[\frac{\cos \phi}{a} (u)_\lambda (v)_\lambda \frac{\partial}{\partial \phi} \left(\frac{[u]_\lambda}{\cos \phi} \right) \right]_{\lambda\phi} + \left[\frac{(v)_\lambda^2}{a} \frac{\partial [v]_\lambda}{\partial \phi} \right]_{\lambda\phi} + \left[\frac{\tan \phi}{a} (u)_\lambda^2 [v]_\lambda \right]_{\lambda\phi} + \left[(\omega)_\lambda (u)_\lambda \frac{\partial [u]_\lambda}{\partial p} \right]_{\lambda\phi} + \left[(\omega)_\lambda (v)_\lambda \frac{\partial [v]_\lambda}{\partial p} \right]_{\lambda\phi} \right) dp \quad (25)$$

where a is the Earth's radius and ω is the vertical velocity in isobaric coordinates. The APE generation and K dissipation terms are defined as:

$$G_Z = \int_{p_t}^{p_b} \frac{[(q)_\lambda]_\phi ([T]_\lambda)_\phi \lambda_\phi}{c_p [\sigma]_\lambda \phi} dp \quad (26)$$

$$G_E = \int_{p_t}^{p_b} \frac{[(q)_\lambda(T)_\lambda]_{\lambda\phi}}{c_p[\sigma]_{\lambda\phi}} dp \quad (27)$$

$$D_Z = - \int_{p_t}^{p_b} \frac{1}{g} [[u]_\lambda[F_\lambda]_\lambda + [v]_\lambda[F_\phi]_\lambda]_{\lambda\phi} dp \quad (28)$$

$$D_E = - \int_{p_t}^{p_b} \frac{1}{g} [(u)_\lambda(F_\lambda)_\lambda + (v)_\lambda(F_\phi)_\lambda]_{\lambda\phi} dp \quad (29)$$

Here, F_λ and F_ϕ represent the zonal and meridional frictional components, respectively, and q is the diabatic heating term, computed as a residual from the thermodynamic equation:

$$\frac{q}{c_p} = \frac{\partial T}{\partial t} + \vec{V}_H \cdot \vec{\nabla}_p T - S_p \omega \quad (30)$$

where $\vec{V}_H \cdot \vec{\nabla}_p T$ represents the horizontal advection of temperature and S_p approximates the static stability, given by:

$$S_p \equiv - \frac{T}{\theta} \frac{\partial \theta}{\partial p} \quad (31)$$

where θ is the potential temperature.

The boundary terms are given by:

$$\begin{aligned} \text{BAZ} = & c_1 \int_{p_1}^{p_2} \int_{\varphi_1}^{\varphi_2} \frac{1}{2[\sigma]_{\lambda\varphi}} \left(2([T]_\lambda)_\varphi (T)_\lambda u + ([T]_{\lambda\varphi})_\varphi^2 u \right)_{\lambda_1}^{\lambda_2} \\ & \times d\varphi dp + c_2 \int_{p_1}^{p_2} \frac{1}{2[\sigma]_{\lambda\varphi}} \left(2[(v)_\lambda(T)_\lambda]_\lambda ([T]_\lambda)_\varphi \cos \varphi + ([T]_\lambda)_\varphi^2 [v]_\lambda \cos \varphi \right)_{\varphi_1}^{\varphi_2} dp \\ & - \frac{1}{2[\sigma]_{\lambda\varphi}} \left([2(\omega)_\lambda(T)_\lambda]_\lambda ([T]_\lambda)_\varphi + [[\omega]_\lambda([T]_\lambda)_\varphi^2]_{\lambda\varphi} \right)_{p_1}^{p_2} \end{aligned} \quad (32)$$

$$\begin{aligned} \text{BAE} = & c_1 \int_{p_1}^{p_2} \int_{\varphi_1}^{\varphi_2} \frac{1}{2[\sigma]_{\lambda\varphi}} [u(T)_\lambda^2]_{\lambda_1}^{\lambda_2} d\varphi dp \\ & + c_2 \int_{p_1}^{p_2} \frac{1}{2[\sigma]_{\lambda\varphi}} \left([(T)_\lambda^2 v]_\lambda \cos \varphi \right)_{\varphi_1}^{\varphi_2} dp \\ & - \left(\frac{[\omega(T)_\lambda^2]_{\lambda\varphi}}{2[\sigma]_{\lambda\varphi}} \right)_{p_1}^{p_2} \end{aligned} \quad (33)$$

$$\begin{aligned} \text{BKZ} = & c_1 \int_{p_1}^{p_2} \int_{\varphi_1}^{\varphi_2} \frac{1}{2g} \left(u [u^2 + v^2 - (u)_\lambda^2 - (v)_\lambda^2] \right)_{\lambda_1}^{\lambda_2} \\ & \times d\varphi dp + c_2 \int_{p_1}^{p_2} \frac{1}{2g} \left([v \cos \varphi [u^2 + v^2 - (u)_\lambda^2 - (v)_\lambda^2]]_{\varphi_1}^{\varphi_2} \right) dp \\ & - \left(\frac{1}{2g} [\omega [u^2 + v^2 - (u)_\lambda^2 - (v)_\lambda^2]]_{\lambda\varphi} \right)_{p_1}^{p_2} \end{aligned} \quad (34)$$

$$\begin{aligned}
\text{BKE} = & c_1 \int_{p_1}^{p_2} \int_{\varphi_1}^{\varphi_2} \frac{1}{2g} \left(u [(u)_\lambda]^2 + (v)_\lambda^2 \right)_{\lambda_1}^{\lambda_2} d\varphi dp \\
& + c_2 \int_{p_1}^{p_2} \frac{1}{2g} \left([v \cos \varphi [(u)_\lambda]^2 + (v)_\lambda^2] \right)_{\varphi_1}^{\varphi_2} dp \\
& - \left(\frac{1}{2g} [\omega [(u)_\lambda]^2 + (v)_\lambda^2] \right)_{p_1}^{p_2}
\end{aligned} \tag{35}$$

where $c_1 = -[a(\lambda_2 - \lambda_1)(\sin \varphi_2 - \sin \varphi_1)]^{-1}$ and $c_2 = -[a(\sin \varphi_2 - \sin \varphi_1)]^{-1}$.
 Lastly, the terms $B\Phi Z$ and $B\Phi E$ are given by:

$$\begin{aligned}
B\Phi Z = & c_1 \int_{p_t}^{p_b} \int_{\varphi_s}^{\varphi_n} \frac{1}{g} ([u]_\lambda \Delta_{EW} \Phi' + \Phi^* \Delta_{EW} u') d\varphi dp \\
& + c_2 \int_{p_t}^{p_b} \frac{1}{g} (\cos \varphi [v]_\lambda \Phi^*)_{\varphi_s}^{\varphi_n} dp \\
& - \frac{1}{g} \left([\omega^* \Phi^*]_{\lambda \varphi} \right)_{p_t}^{p_b}
\end{aligned} \tag{36}$$

$$\begin{aligned}
B\Phi E = & c_1 \int_{p_1}^{p_2} \int_{\varphi_1}^{\varphi_2} \frac{1}{g} ((u)_\lambda (\Phi)_{\lambda_\lambda})_{\lambda_1}^{\lambda_2} d\varphi dp \\
& + c_2 \int_{p_1}^{p_2} \frac{1}{g} \left([(v)_\lambda (\Phi)_{\lambda_\lambda}]_\lambda \cos \varphi \right)_{\varphi_1}^{\varphi_2} dp \\
& - \frac{1}{g} \left([(\omega)_\lambda (\Phi)_\lambda]_{\lambda \varphi} \right)_{p_1}^{p_2}
\end{aligned} \tag{37}$$

Here, $\Phi^* = [\Phi]_\lambda - [\Phi]_{\lambda \varphi}$, $\omega^* = [\omega]_\lambda - [\omega]_{\lambda \varphi}$, $\Phi' = \Phi - [\Phi]_\lambda$, $u' = u - [u]_\lambda$, and $\Delta_{EW} X = X|_{\lambda_2} - X|_{\lambda_1}$.

Declarations

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Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Data Availability

The cyclone tracks used in this study were obtained from the *Atlantic Extratropical Cyclone Tracks Database*, available at <https://doi.org/10.17632/kwcvfr52hp>.
 4. The archived article inputs and scripts are available at https://github.com/daniloceano/energetic_patterns_cyclones_south_atlantic. The corrected calculation, validation products, statistics, figures, and provenance used for this correction

are maintained at <https://github.com/daniloceano/lec-climatology-rerun>. The source code of the **Cyclophaser** package is available on PyPI at <https://pypi.org/project/cyclophaser/>, and the corrected **LorenzCycleToolkit** is available at <https://github.com/daniloceano/LorenzCycleToolkit>.

Author contributions

Conceptualization: Danilo Couto de Souza, Pedro Leite da Silva Dias, Ricardo de Camargo, Carolina Barnez Gramscianinov; Methodology: Danilo Couto de Souza, Pedro Leite da Silva Dias, Ricardo de Camargo; Formal analysis and investigation: Danilo Couto de Souza, Pedro Leite da Silva Dias; Writing - original draft preparation: Danilo Couto de Souza; Writing - review and editing: Danilo Couto de Souza, Pedro Leite da Silva Dias, Ricardo de Camargo, Carolina Barnez Gramscianinov; Funding acquisition: Pedro Leite da Silva Dias, Ricardo de Camargo; Resources: Pedro Leite da Silva Dias, Ricardo de Camargo; Supervision: Pedro Leite da Silva Dias, Ricardo de Camargo.

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1038 Supplementary information

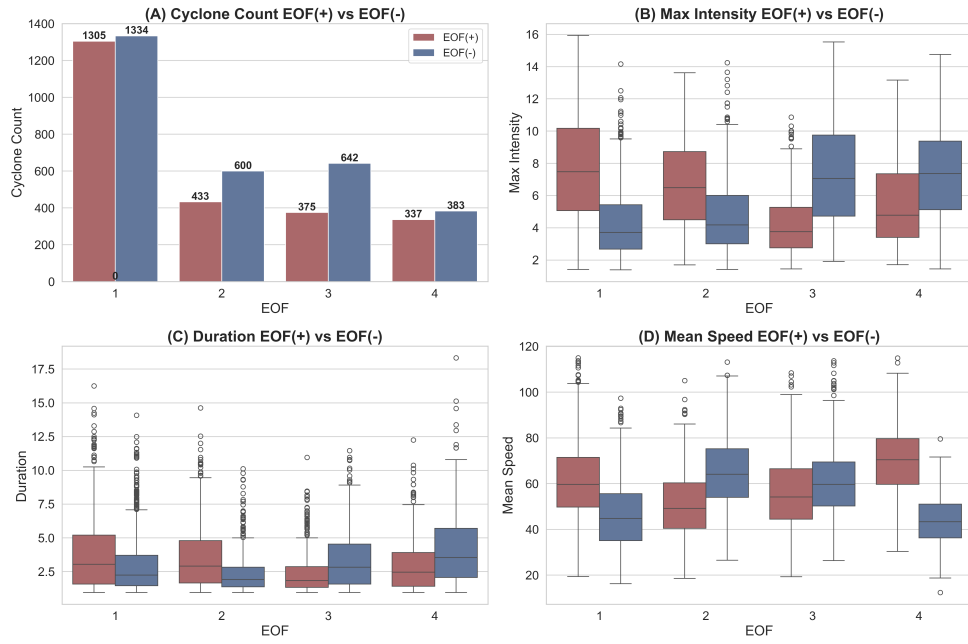


Fig. S1 Statistics for the cyclones classified as EOF(+) (red) and EOF(-) (blue). (A) Cyclone counts; (B) maximum intensity, expressed by the *minimum* central relative vorticity at 850 hPa; (C) total life-cycle duration (days); and (D) mean translational speed (km h^{-1}).

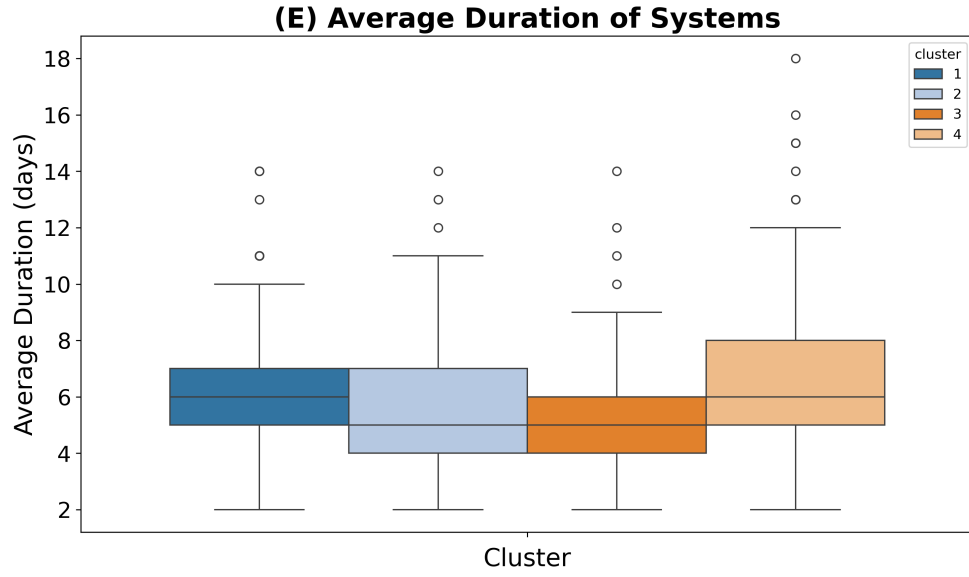


Fig. S2 Legacy four-group duration panel retained as a placeholder. The canonical corrected analysis contains five clusters; this graphic must be replaced by a five-group duration panel before publication.

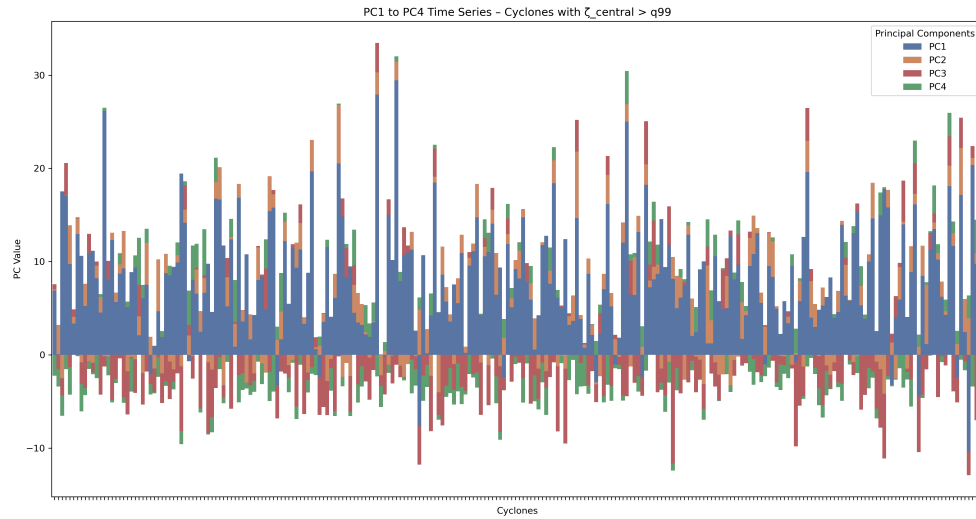


Fig. S3 Stacked time series of the first four principal components (PC1–PC4) for each intense cyclone ($\zeta_{850, \text{max}} > q_{90}$). Positive values are plotted upward and negative values downward, so the coloured bars represent the signed contribution of each EOF mode to the energetics of every system. The x -axis lists the individual cyclones (sorted arbitrarily), while the y -axis gives the PC amplitude (dimensionless).

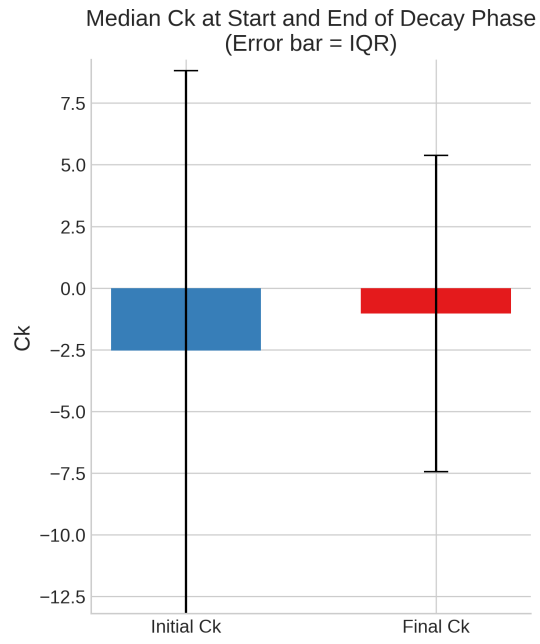


Fig. S4 Median barotropic-conversion term (C_k) at the start (blue) and end (red) of the decay phase, with the error bars representing the inter-quartile range.